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Mechanics of Long-Context Hybrid Models
Part 1.1: From Hybrid Attention to Hybrid Position

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Abstract

The architectural design of mainstream open-source Large Language Models (LLMs) is shifting from traditional full-softmax-attention-only models to hybrid models, which combine different attention modules across layers or heads and introduce different position embeddings to improve long-context efficiency and performance. To this end, we propose **Mechanics of Long-Context Hybrid Models**, aiming to answer why hybrid models are effective and how to design them more effectively.

As Part 1.1 of this series, we begin with hybrids of full attention and either sliding-window attention (SWA) or gated variants of linear attention (LA), represented by GLA and GDN. We first observe a **Seesaw Effect in Context Extension**: SWA hybrids perform better in length extrapolation, while LA hybrids perform better after context extension. We attribute this to their different position biases. We find that SWA hybrids suffer from a **Short-Context Learning Trap**, Short-Window Weariness, and Long-Window Laziness, requiring extended windows to enhance performance in continual long-context pretraining. In contrast, we summarize the **Matthew Effect of Hybrid Position Extrapolation** and propose **Sliding-Window Linear Attention**, achieving 16× training-free length extrapolation while maintaining 100% retrieval accuracy.

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Code Repository: https://github.com/OpenMOSS/Hybrid-Mechanics

Takeaway List

T1 From Hybrid Attention to Hybrid Position. Hybrid models improve long-context performance and length extrapolation through hybrid position, combining NoPE with other position biases.

T2 Seesaw Effect in Context Extension of SWA Hybrid. The advantage of SWA-NoPE hybrids over LA-NoPE hybrids in long-context performance is reversed after long-context pretraining, especially for layer-wise hybrids, due to their **Short-Context Learning Trap in Context Extension**.

T3 No Free Lunch Effect in Length Extrapolation of LA Hybrid. LA-NoPE hybrids are weaker than SWA-NoPE hybrids in extrapolation, but stronger within the training context.

T4 Tidal Effect in Collaboration of Hybrid Position. Effective collaboration of hybrid position relies on a minority of high-hit-rate NoPE attention with coarse aggregation and a majority of low-entropy position-biased attention (i.e., RoPE attention and gated linear attention) for noise reduction.

T5 Short-Window Weariness and Long-Window Laziness of SWA Hybrid. In length extrapolation, SWA hybrids with short window size perform better, but in context extension, SWA hybrids with sliding window extended to a larger size overcome the short-context trap better.

T6 Matthew Effect in Length Extrapolation of Hybrid Position. Effective extrapolation of hybrid models requires SWA in position-biased attention and enhanced aggregation in NoPE attention.

1 Introduction

The architectural design of mainstream open-source Large Language Models (LLMs) [1–6] is shifting from traditional full-softmax-attention-only models [7–10] to hybrid models [11–19], which combine different attention modules across layers or heads and introduce different position embeddings to improve long-context efficiency and performance [20–22]. To this end, we propose **Mechanics of Long-Context Hybrid Models**. Through systematic training, evaluation, mechanism analysis, and closed-loop validation, we analyze how different attention modules inside a hybrid architecture collaborate and how these collaborations affect length extrapolation and long-context performance, aiming to answer "**why hybrid architectures are effective**" and "**how to design them more effectively**" for long-context LLMs.

- **Why Hybrid?** Hybrid models have become an important paradigm in LLM design [12, 15, 16]. Compared with traditional RoPE-based full-attention-only models [23, 24], hybrid models not only offer higher computational and memory efficiency [9, 10, 19], but also provide better performance in retrieval, state tracking, language modeling, and length extrapolation [21, 22, 25–28].
- **Why Long-Context?** LLMs need to support longer context [14, 19, 29–31], and long context is one of the most important application scenarios for current hybrid models [21, 22, 25]. At the same time, million-token-level contexts in the agentic era impose strong constraints on model architectures [14, 15, 19]. Relying solely on full attention is unrealistic, while any single efficient attention is unreliable [21, 32, 33]. This makes hybrid models one of the most worth-studying fields in long-context LLMs [20, 22, 34, 35].
- **Why Mechanics?** There has already been substantial work discussing hybrid models from the perspectives of downstream task evaluation, scaling curves, or system efficiency [20, 22, 25, 34, 35], but these results often remain at the level of comparing the phenomenon of "which model is better" [22, 34]. We instead focus on how different attention modules inside a hybrid model interact, collaborate, and affect length extrapolation [36] during inference and context extension during training. This is why we use Mechanics to summarize this series of works.

On account of this long journey involving comparisons across multiple hybrid modules, hybrid paradigms, and training scenarios, we proceed step by step. This paper is Part 1.1 of the whole series.

- **In Part 1**, we first discuss the hybridization of memory-constant sliding-window attention (SWA) or linear attention (LA) with full attention, which is also one of the most widely used classes of hybrid models at present [12, 13, 15–17, 22]; in subsequent work, we will continue to explore hybrid models such as sparse-attention hybrids [10, 27, 37–39] and compressed-attention hybrids [19, 40].
- **In Part 1.1**, we start from the perspective of hybrid position, focusing on the length extrapolation (i.e. training-free generalization beyond the pretraining context length) and context extension (i.e. long-context continual pretraining that expands context length) performance of RoPE-based full-attention models, RoPE-NoPE hybrid-position full-attention models, SWA hybrid models, and LA hybrid models represented by GLA/GDN [16, 32, 33]. In this paper, we focus on the long-context language modeling [41, 42], retrieval [43, 44], and state-tracking abilities [45] in pretraining; in subsequent work, we will continue to explore more advanced abilities in post-training and multi-modal training.

Regarding the results obtained in Part 1.1, we give a complete Takeaway List on the first page of the paper. Specifically, the contributions of this paper can be summarized as follows:

- We compare RoPE-based full attention, RoPE-NoPE hybrid, SWA hybrid, and GLA/GDN hybrid, both layer-wise and head-wise, across model scales of 376M, 776M, and 1B through long-context retrieval, state-tracking, and language-modeling performance under length extrapolation in short-context pretraining and context extension after long-context continual pretraining.
- **On hybrid position**, we find that hybrid models improve long-context performance and length extrapolation by combining NoPE with other positional biases, i.e., RoPE attention and gated linear

attention (**Takeaway 1**). They occupy distinct functional regions in the entropy-hit-rate scatter diagram, where the boundaries shift with the hybrid ratio (**Takeaway 4**, **Tidal Effect**).

- **On SWA hybrids**, we demonstrate that although SWA hybrids exhibit clear advantages in length extrapolation, they suffer from the short-context learning trap in context extension (**Takeaway 2, Seesaw Effect**). To address this, we improve SWA hybrids by enlarging the window size and applying the LongCE loss, and further summarize the underlying short-window weariness (**Takeaway 5**).
- **On LA hybrids**, we find that although LA hybrids perform better in long-context continual pretraining, they are weaker in length extrapolation (**Takeaway 3**, **No Free Lunch Effect**). Through our extended attention-entropy analysis, we observe that LA and RoPE attention exhibit similar patterns in the entropy-hit-rate diagram, and both can be viewed as position-biased attention. Based on this, we propose **Sliding-Window Linear Attention**, achieving 16× length extrapolation with 100% retrieval accuracy, and summarize the underlying **Matthew Effect** in hybrid position extrapolation (**Takeaway 6**).

2 Related Work

2.1 The Rise of Hybrid Models

LLM architecture design has shifted from traditional full-attention-only models to hybrid models [11–19], combining attention modules with different forms, complexities, and position embeddings to mix information along the sequence dimension [20, 21]. For example, Qwen-3.5 [16, 46], Kimi-K3 [15, 21], and OLMo-Hybrid [22] mix full softmax attention (e.g., GQA [8] and MLA [9]) or sparse attention (e.g., DSA [10]) with linear attention (e.g., GDN [33] and KDA [21]). Gemma4 [13], Inkling [47], and Mimo-v2 [17] mix full softmax attention with sliding-window softmax attention. GLM-5.3 [18, 48] and MiniMax-M3 [37, 49] adopt sparse-attention hybrids [10, 38, 39], while DeepSeek-v4 [19] uses compressed-attention hybrids [40] with different compression intensity. Hybrid attention often comes with hybrid positional biases and has position-embedding designs that differ from those in traditional models. For example, Jamba [11], Kimi-K3 [15], SWAN-GPT [25], and RNoPE [28] also show that full-attention layers in hybrid models do not necessarily require explicit position embeddings like RoPE [23].

These hybrid models can be more computationally and memory efficient than traditional models, while achieving downstream performance close to or even better than traditional full-attention models. As a result, hybrid models have received increasing attention [15, 20, 22, 34]. Among them, hybrids with fixed memory complexity, especially linear attention (LA) and sliding-window attention (SWA) hybrids, after the early exploration [50**?** , 51], have since been adopted by LLMs including Jamba [11], GPT-OSS [12], MiniMax-01 [14], Qwen [16, 46], Kimi [15, 21], Nemotron-H [52], and MiniCPM-SALA [53]. They have also inspired fine-tuning attempts [54–57], and refined efficiency optimizations [26, 58]. Therefore, we start from this class of hybrids to study why hybrid models are effective and how to design them more effectively.

2.2 Long-Context Performance of Hybrid Models

Beyond efficiency, previous work also suggests that hybrid models can bring performance gains within the training context length [15, 22, 28]. Early works [59, 60] show that hybrid models can combine the advantages of traditional softmax attention in retrieval with the advantages of linear attention in state tracing. Kimi-K3 [15] and OLMo Hybrid [22] further verify at larger scales that LA hybrids can improve model expressivity and pretraining scaling efficiency. In addition to layer-wise hybrids, Hymba [61] and HydraHead [35] also verify the effectiveness of head-wise hybrids.

Beyond empirical validation, some works have begun to analyze SWA and LA hybrids theoretically. For example, Wang et al. [20] suggests that a ratio around 3:1 suits hybrid models, while Wu et al. [62] points out that retrieval is mainly handled by full attention, whereas SWA or LA affects the training trajectory. Although these works explain the benefits of hybrid models, there is still a lack of mechanistic analysis of how attention modules cooperate based on their different positional inductive biases in length extrapolation and context extension. Based on this, can we combine the strengths and weaknesses of different attention modules in their positional-bias designs and derive new hybrid strategies that advance length extrapolation and context extension? These are the questions we study in this paper.

2.3 Extrapolation based on Hybrid Models

Besides improving long-context performance within the training length, hybrid models have also been shown to improve length extrapolation, or length generalization [36]. For example, iRoPE [63], RNoPE [28], and SWAN-GPT [25] use RoPE only in some layers and restrict the attention window of these layers during extrapolation, while keeping global position awareness only in NoPE layers, achieving strong extrapolation performance. This hybrid position design has also proven to improve retrieval performance within the training length [28]. Similarly, in LA hybrids, Jamba [11], Kimi-K3 [15], and HypeNet [55] find that RoPE can also be removed from full-attention layers, which also helps direct length generalization [36].

Although LA hybrids do not use explicit positional embeddings, this does not mean there is no positional bias. Linear attention also contains various position designs, some of which have already been used to improve traditional softmax attention and build stronger position embeddings or attentions [64], such as FoX [65], DeltaFormer [66], and PaTH [67]. Accordingly, we can extrapolate LA hybrids in the same way as RoPE-NoPE hybrids by limiting position-biased attention to local windows while strengthening global perception in modules without positional bias, which is also one of the contributions in this paper.

3 Observation

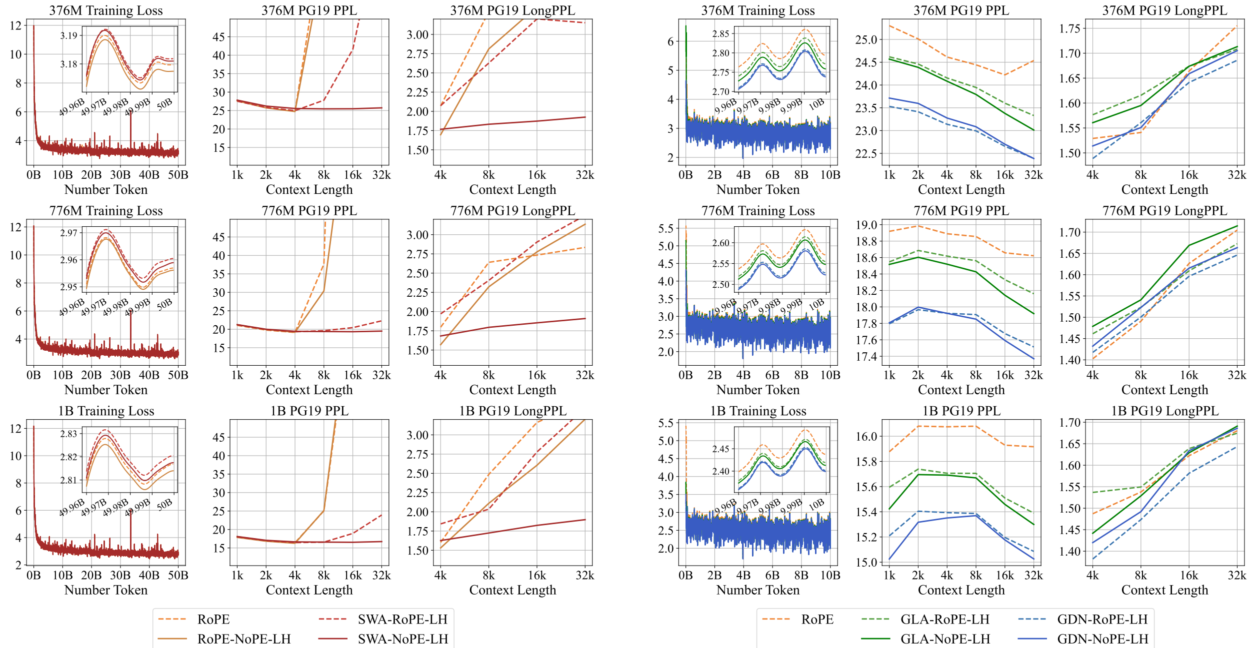
3.1 Short-Context Training

We conduct experiments at 376M, 776M, and 1B model sizes, with the configuration detailed in Table 2 in Appendix A. All models are pretrained on the DCLM-Baseline-1.0 corpus [68] in BFloat16 precision with a 4k context length using DeepSpeed [69] on 8 NVIDIA H200 140 GB GPUs, accelerated by FlashAttention2 [70] and FlashLinearAttention [71]. We also verify important conclusions by pretraining on a 3B model size with 16 NVIDIA H200 GPUs. For each size, we use a batch size of 1M tokens and pretrain for 50B tokens. We use the AdamW [72] optimizer with weight decay 0.1, a maximum learning rate of 1e-3, and a cosine annealing scheduler. We use the first 0.5B tokens for warm-up, and the learning rate ends at 0.

We use the OpenCompass [73] framework for evaluation, focusing on the LLMs' long-context task performance, including performance within the training length and extrapolation beyond the training length. We use the PG19 dataset [41] for perplexity (PPL) and longPPL [42] evaluation, and use RULER [44] and BABILong [45] to evaluate retrieval and state-tracking abilities, respectively. In addition, we also report the average scores within 4k and beyond 4k as references for in-domain and out-of-domain performance.

We abbreviate layer-wise hybrid as LH and head-wise hybrid as HH, and use RoPE and NoPE to denote full-attention layers or heads using the corresponding position embedding. For example, RoPE denotes the pure full-RoPE full-attention model. RoPE-NoPE-LH denotes a full-attention model where RoPE and NoPE are mixed by layer, with a default ratio of 3:1 [28]. SWA-RoPE-LH denotes a model where SWA and RoPE full attention are mixed by layer with ratio 3:1 [16, 20]; correspondingly, SWA-NoPE-LH denotes a model where SWA and NoPE full attention are mixed by layer [25]. Layer-wise hybrids with GLA or GDN and head-wise hybrids are named similarly. In this paper, when we refer to gated linear-attention hybrids, we specifically mean the GLA/GDN family. We set the window size of SWA to 128, following GPT-OSS [12] and Mimo-v2.5 [17], and set the rotary base of SWA and RoPE full attention to 10000 by default [23], while GLA and GDN follow their original papers [32, 33, 71]. We only use MHA, since GDN does not support GQA.

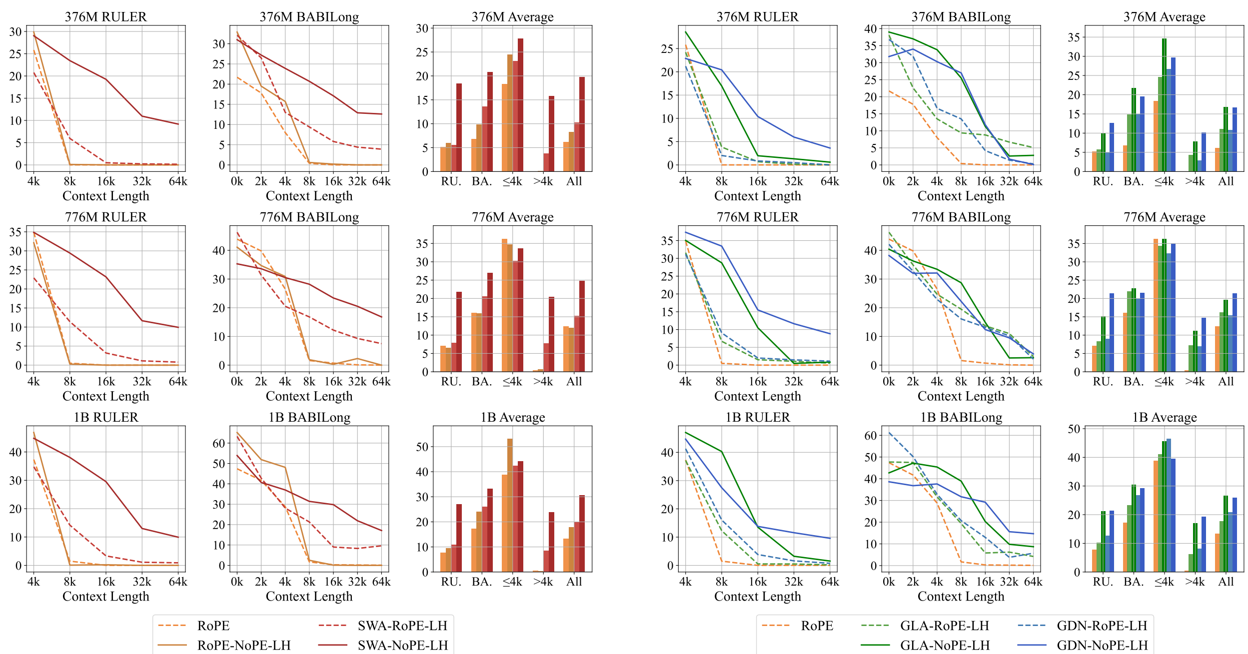
Figure 1 and Figure 2 show the results of layer-wise hybrids, while Figure 3 and Figure 4 show those of head-wise hybrids. Compared with the RoPE-only full attention model, the hybrid models exhibit lower training loss and perplexity within the training length, even for RoPE-NoPE hybrid full attention models [28]. In SWA hybrids and LA hybrids, removing the positional embedding in the full attention layer still enhances long-context performance and length extrapolation [11, 25], in both layer-wise and head-wise hybrids.



(a) Results on Full Attention and SWA Hybrid.

(b) Results on Full Attention and GLA/GDN Hybrid.

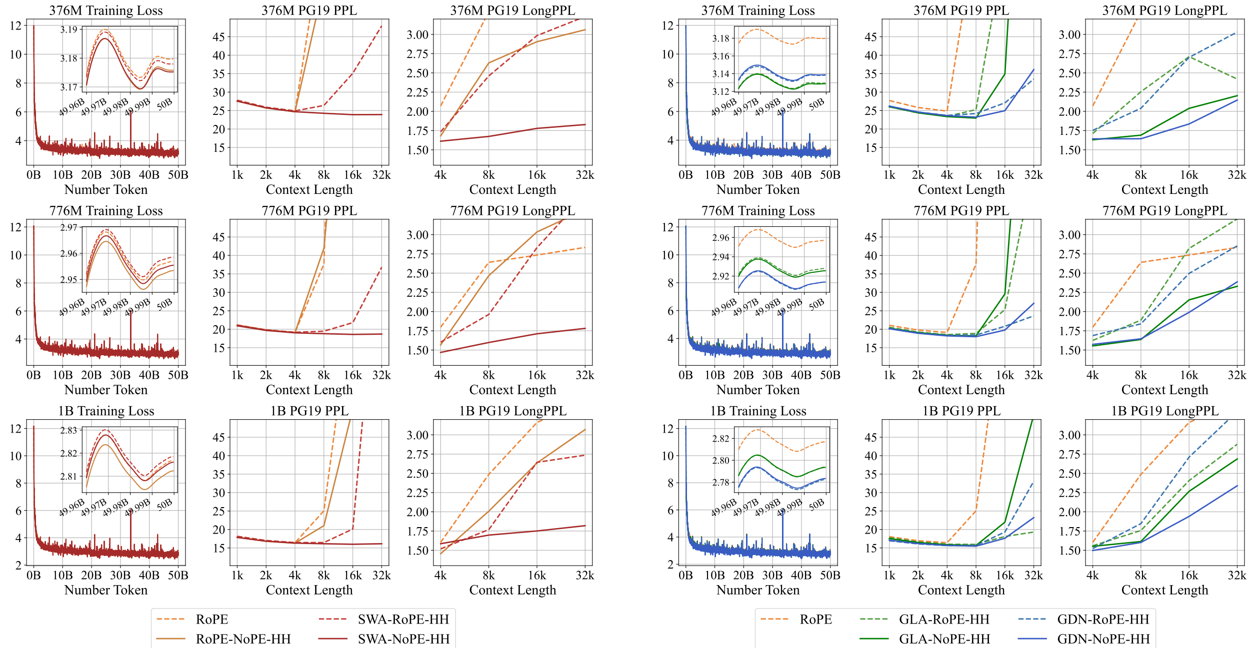
Figure 1 Comparison of training loss, perplexity (PPL), and LongPPL among different layer-wise hybrid models after 4k context length pretraining. SWA/GLA/GDN-NoPE hybrids achieve stronger long-context language modeling.



(a) Results on Full Attention and SWA Hybrid.

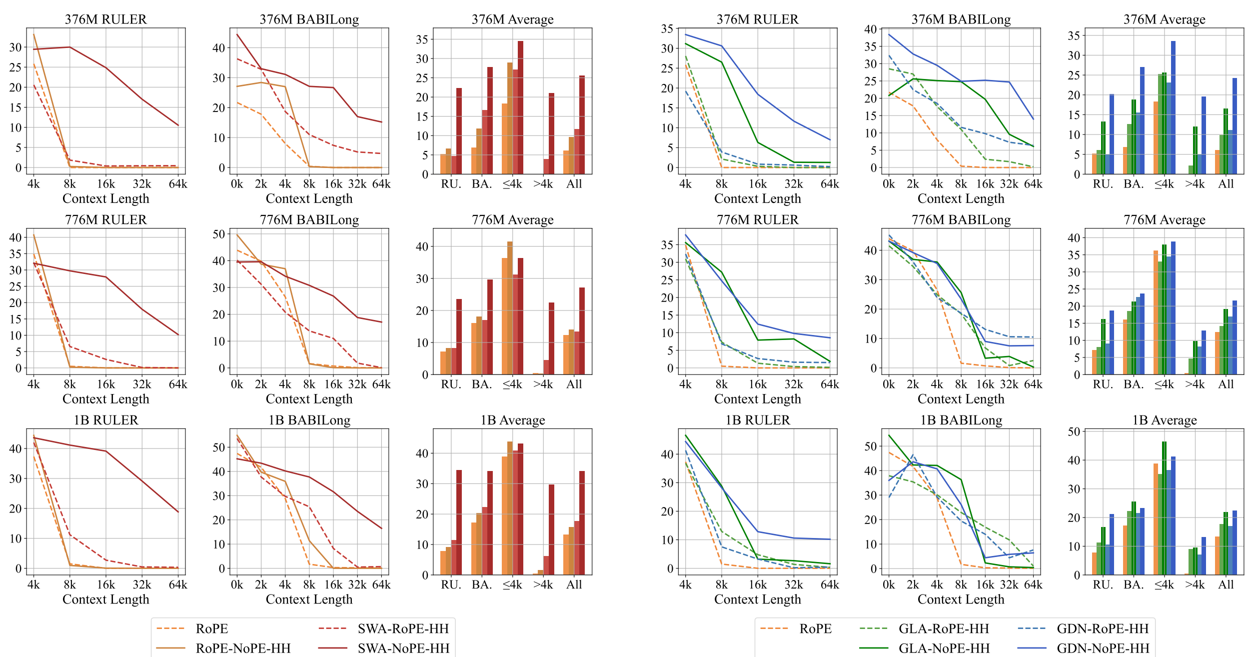
(b) Results on Full Attention and GLA/GDN Hybrid.

Figure 2 Comparison of long-context performance under RULER and BABILong benchmarks among different layer-wise hybrid models after 4k context length pretraining. SWA/GLA/GDN-NoPE hybrids achieve both stronger long-context downstream performance within and beyond the training context length.



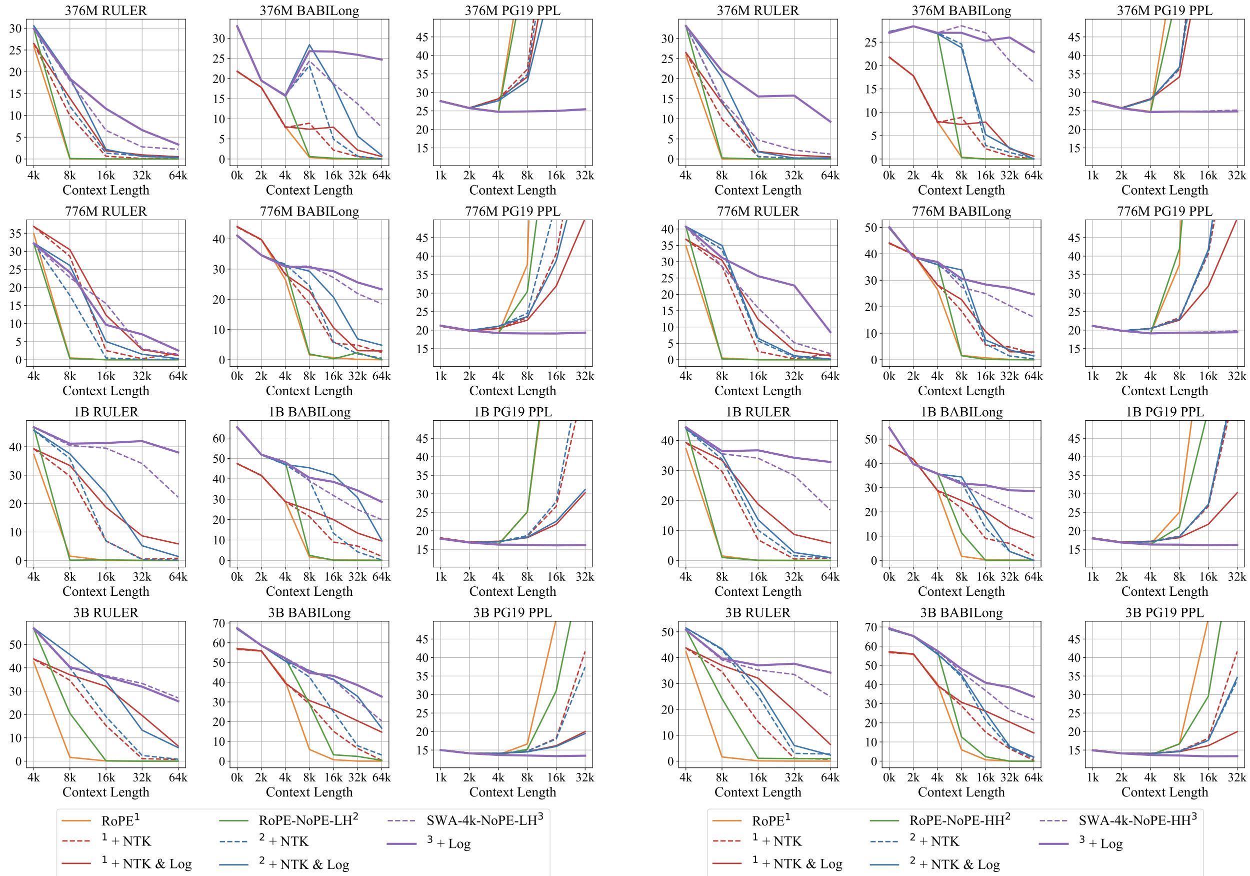
(a) Results on Full Attention and SWA Hybrid. **(b)** Results on Full Attention and GLA/GDN Hybrid.

Figure 3 Comparison of training loss, perplexity (PPL), and LongPPL among different head-wise hybrid models after 4k context length pretraining. SWA/GLA/GDN-NoPE hybrids achieve stronger long-context language modeling.



(a) Results on Full Attention and SWA Hybrid. **(b)** Results on Full Attention and GLA/GDN Hybrid.

Figure 4 Comparison of long-context performance under RULER and BABILong benchmarks among different head-wise hybrid models after 4k context length pretraining. SWA/GLA/GDN-NoPE hybrids achieve both stronger long-context downstream performance within and beyond the training context length.



(a) Results on Layer-wise Hybrid Model (LH). **(b)** Results on Head-wise Hybrid Model (HH).

Figure 5 Comparison of long-context performance between RoPE-only and RoPE-NoPE hybrid full-attention models. Both layer-wise and head-wise RoPE-NoPE hybrids have better long-context performance and can achieve effective length extrapolation directly or under log-scaled NoPE extrapolation.

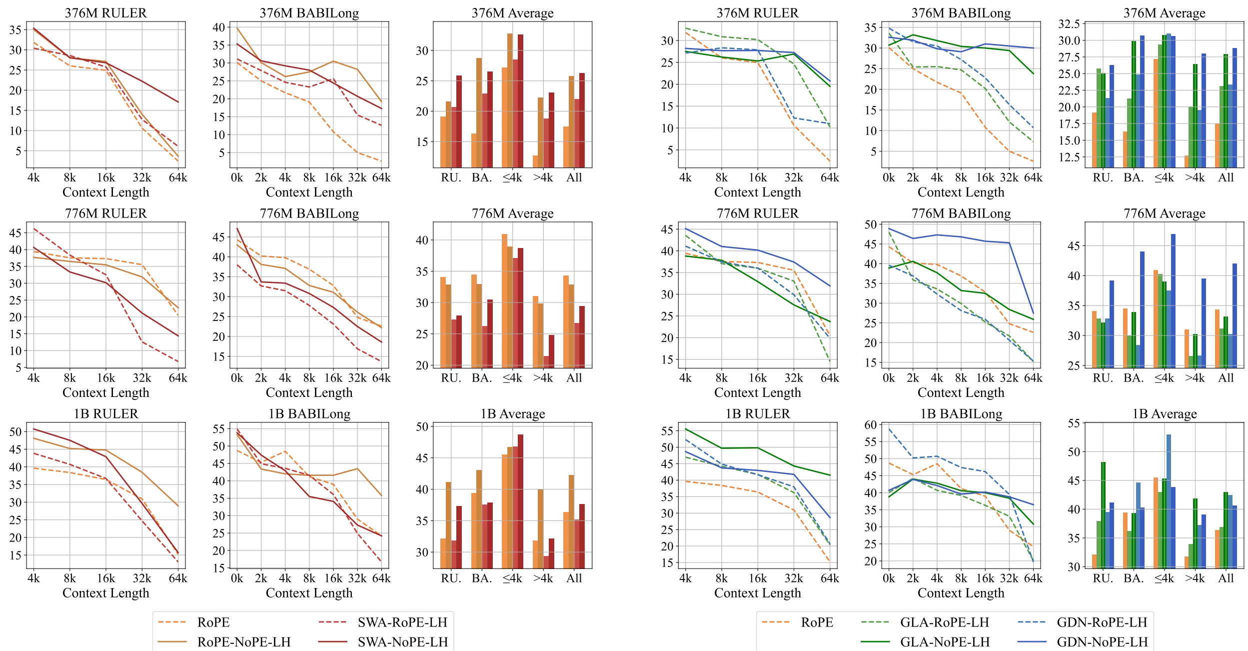
We also verify the effectiveness of RoPE-NoPE extrapolation in an SWA-NoPE hybrid manner [25, 28] rather than positional interpolation [74–77]. For RoPE attention, we retain only the attention window of the nearest pretraining context length, 4k. For NoPE attention, we use a log-scaled attention extrapolation, increasing the attention temperature to prevent attention entropy from increasing as the context grows, with the log base equal to the pretraining context length [78, 79]. As shown in Figure 5, this hybrid extrapolation method significantly outperforms RoPE or RoPE-NoPE hybrids using Dynamic NTK interpolation [75, 79].

Besides, since the gate mechanism of linear attention can also be regarded as a data-dependent position embedding [65–67], we argue that hybrid position bias is key to achieving better performance on long contexts in the hybrid models, which is the **Takeaway 1** in this paper, **From Hybrid Attention to Hybrid Position**. We will provide a detailed analysis in Section 4 and Section 6.

Takeaway 1: From Hybrid Attention to Hybrid Position

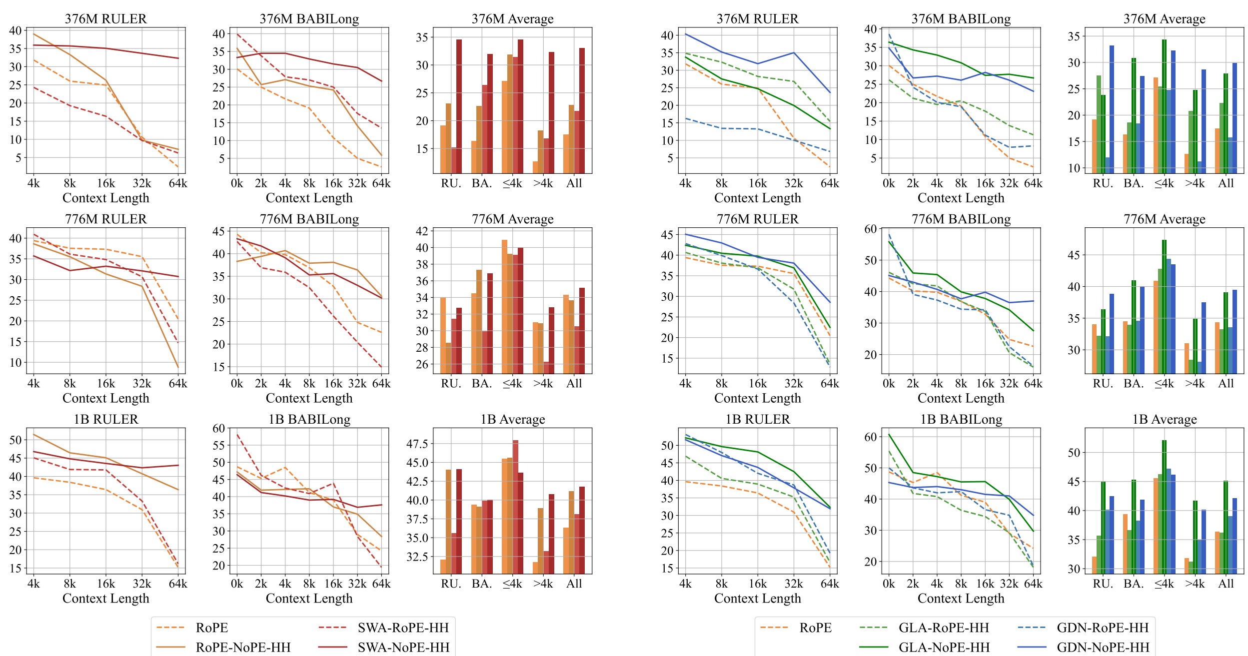
Hybrid models improve long-context performance and length extrapolation through a hybrid position of NoPE and other position biases. Even in RoPE-NoPE hybrid models, training-free effective length extrapolation is achieved for LLMs pretrained with full attention.

3.2 Long-Context Training



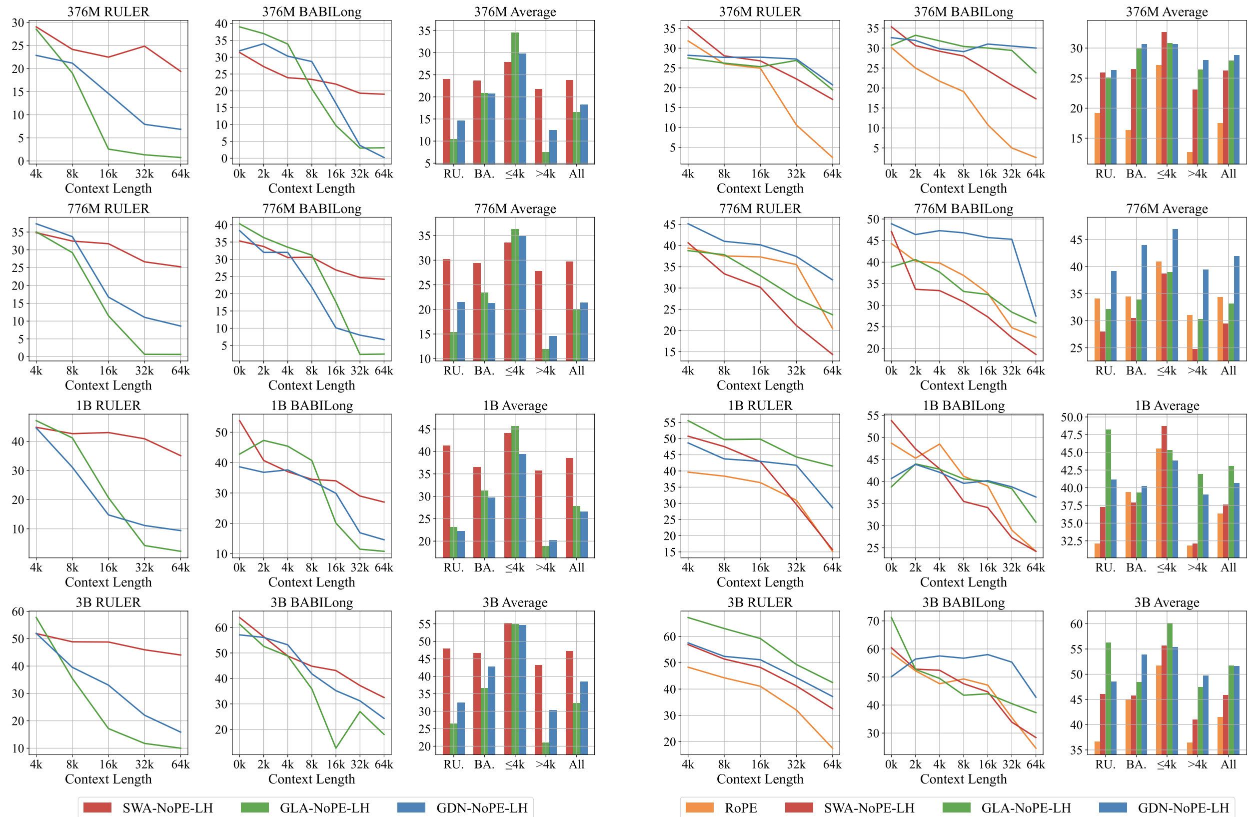
(a) Results on Full Attention and SWA Hybrid. **(b)** Results on Full Attention and GLA/GDN Hybrid.

Figure 6 Comparison of long-context performance under RULER and BABILong benchmarks among different layer-wise hybrid models after 32k context length continual pretraining.



(a) Results on Full Attention and SWA Hybrid. **(b)** Results on Full Attention and GLA/GDN Hybrid.

Figure 7 Comparison of long-context performance under RULER and BABILong benchmarks among different head-wise hybrid models after 32k context length continual pretraining.



(a) Performance of log-scale NoPE extrapolation. **(b)** Performance after long-context continual pretraining.

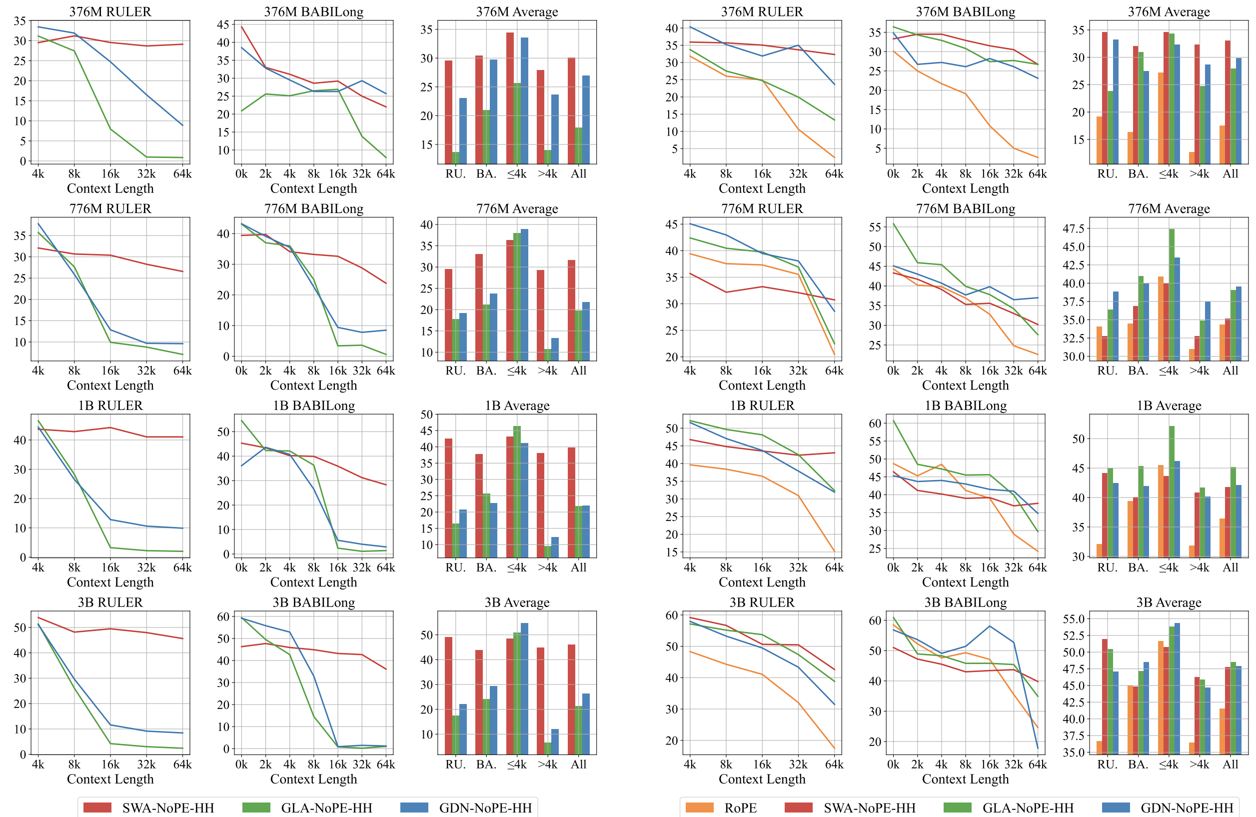
Figure 8 Comparison of the long-context performance of the SWA/GLA/GDN-NoPE layer-wise hybrid models under log-scale NoPE extrapolation after short-context pretraining (left) and after long-context continual pretraining (right). The results of SWA hybrids exhibit a remarkable seesaw effect, as detailed in Takeaway 2.

We further conduct long-context continual pretraining. We use holistic long documents from the high-quality long-context corpus LongWanjuan [80], and continue training at a 32k context length. We follow the data mixture ratios used in the original paper [24, 80], and increase the rotary base from 10000 to 500000 for RoPE full-attention to support contexts of around 64k tokens [79, 81]. We use a 0.5M batch size and continue training for 5B tokens. We use the AdamW [72] optimizer with 0 weight decay, a peak learning rate of 1e-3, and a cosine annealing scheduler. We use the first 1B tokens for warm-up, and finally decay the learning rate to 0. Figure 6 shows results on layer-wise hybrids, while Figure 7 shows those on head-wise hybrids.

Notably, there exists a very interesting phenomenon. SWA hybrids, which perform best in length extrapolation, underperform the LA hybrids after context extension, as shown in Figure 8 and Figure 9, and may even fail to surpass their performance under log-scaled NoPE extrapolation, which is more remarkable in layer-wise hybrids. We call this phenomenon of SWA hybrids being outperformed by LA hybrids in long-context pretraining the **Seesaw Effect of Context Extension in SWA Hybrid**, which is **Takeaway 2** in this paper. We

Takeaway 2: Seesaw Effect in Context Extension of SWA Hybrid

For the SWA-NoPE hybrid model, the advantage in long-context performance over the LA-NoPE hybrid is inverted after long-context continual pretraining. Moreover, this phenomenon is more remarkable in the SWA-NoPE layer-wise hybrid model.



(a) Performance of log-scale NoPE extrapolation.

(b) Performance after long-context continual pretraining.

Figure 9 Comparison of the long-context performance of the SWA/GLA/GDN-NoPE head-wise hybrid models under log-scale NoPE extrapolation after short-context pretraining (left) and after long-context continual pretraining (right). The results of SWA hybrids exhibit a remarkable seesaw effect, as detailed in Takeaway 2.

will provide an in-depth analysis in Section 5 and try to enhance the context extension of SWA hybrids.

In contrast, LA hybrids benefit more from context extension during long-context continual pretraining, but lag behind SWA hybrids in length extrapolation. In other words, compared with SWA hybrids, LA hybrids exhibit stronger fitting capability within the training context length, yet weaker generalization beyond it. This trade-off is consistent with the "no free lunch" principle in neural architecture design, leading to **Takeaway 3** in this paper, **No Free Lunch Effect in LA Hybrids**. We will answer why LA hybrids have better long-context transfer capabilities in Section 6 and try to enhance the length extrapolation of LA hybrids.

Takeaway 3: No Free Lunch Effect in Length Extrapolation of LA Hybrid

The GLA/GDN-NoPE hybrid model shows its disadvantages in extrapolation (OOD) compared to the SWA-NoPE hybrid model and advantages within the training context (in-domain). Particularly,

- GLA/GDN-NoPE underperforms SWA-NoPE under direct extrapolation and log-scaled NoPE extrapolation after short-context pretraining, while
- GLA/GDN-NoPE outperforms SWA-NoPE within the training context length in both short- and long-context pretraining.

4 Discussion on Hybrid Position

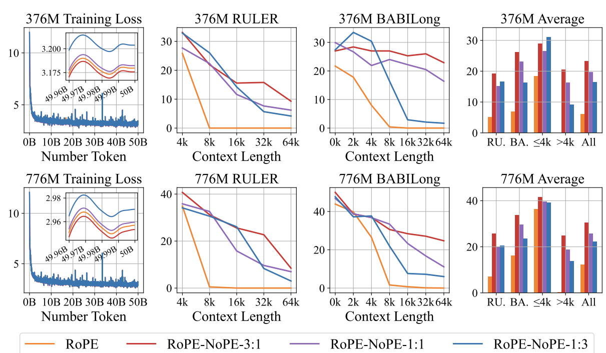
The effectiveness of hybrid models raises thought-provoking questions: why do RoPE/NoPE-only full attention, sliding-window attention, and linear attention each have shortcomings, but hybrid models can achieve better long-context performance within and beyond the training context length? What specific roles do NoPE and other position-biased attention play in the same model?

4.1 Ratio Ablation and Noise Analysis

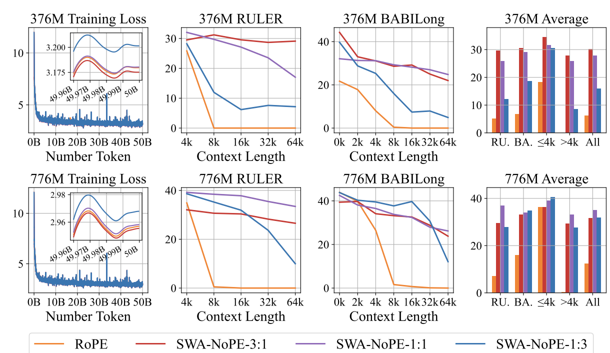
To answer these questions, we first analyze NoPE head-wise hybrid models with different NoPE ratios. We choose head-wise hybrids because they let NoPE and other attention types collaborate in parallel within the same layer, making them better suited for role attribution than layer-wise hybrids, and avoiding the impact of layer order and functional differences at different depths.

We compare RoPE-NoPE-HH, SWA-NoPE-HH, GLA-NoPE-HH, and GDN-NoPE-HH with different hybrid ratios, 3:1, 1:1, and 1:3, using the same setup for pretraining on model sizes of 376M and 776M. We only conduct the first stage of pretraining with a context length of 4k, and apply a sliding window with a window size of 4k for RoPE attention in RoPE-NoPE-HH [25, 28], as well as a log attention scale for NoPE attention in all hybrids to achieve better length extrapolation [78, 79]. The results are shown in Figure 10. The results indicate that a 3:1 RoPE-NoPE ratio performs best within and beyond the training length in most cases. While NoPE attention is key to improving the hybrid model's long-context performance and extrapolation capability, more NoPE attention actually weakens the extrapolation effect.

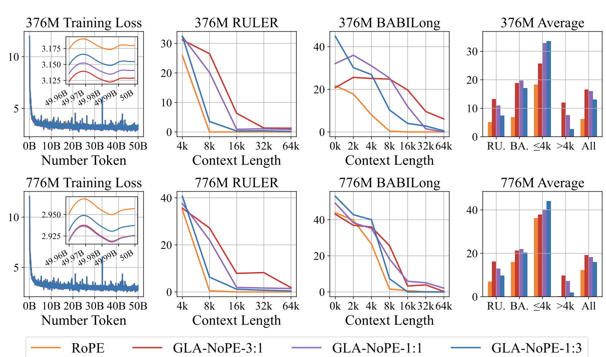
Particularly, we conduct noise experiments on the 1:1 NoPE head hybrids. We add Gaussian noise to NoPE heads or other position-biased heads and observe which part of the noise has a greater impact on downstream



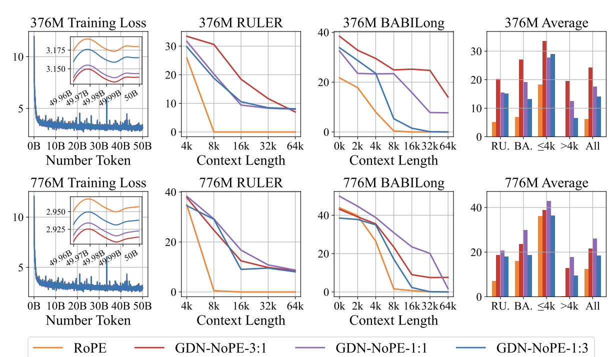
(a) Results on RoPE-NoPE hybrids.



(b) Results on SWA-NoPE hybrids.

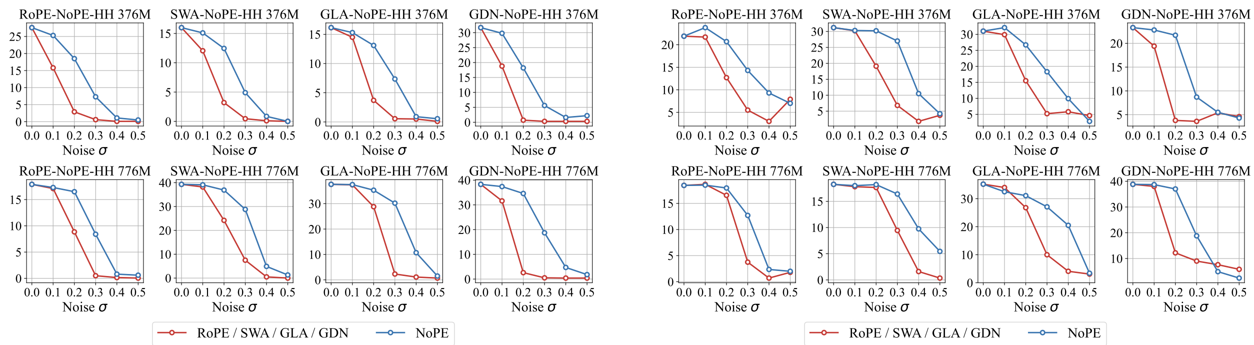


(c) Results on GLA-NoPE hybrids.



(d) Results on GDN-NoPE hybrids.

Figure 10 Comparison of training loss, RULER, and BABILong among different mixing ratios in head-wise hybrid models.



(a) Results on RULER Benchmark.

(b) Results on BABILong Benchmark.

Figure 11 Results of noise experiments on RULER and BABILong benchmark. Adding Gaussian noise to RoPE full attention or SWA/GLA/GDN degrades long-context performance more severely than the same perturbation applied to NoPE full attention in the corresponding hybrid models.

tasks such as RULER and BABILong in 4k context length as the variance increases. Surprisingly, as shown in Figure 11, adding noise to the position-biased heads, including RoPE/SWA/GLA/GDN heads, has a greater impact on downstream long-context tasks. This contradicts the intuition that NoPE is more responsible for global information retrieval, tending to dominate long-context performance, prompting us to investigate further: what role does NoPE full attention play in the hybrid models, and why could adding only a small amount of NoPE attention improve the performance of the original RoPE model?

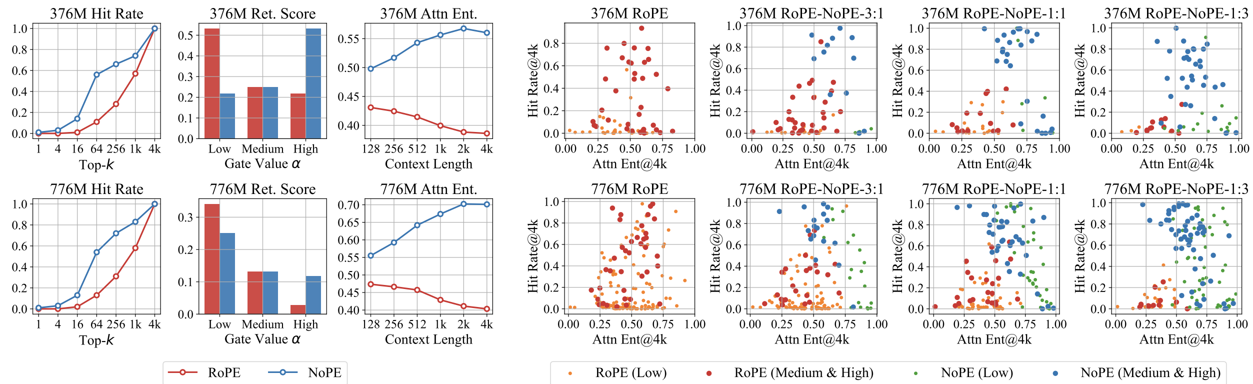
4.2 Attention Analysis of Hybrid Position

To analyze the role of NoPE full attention in the hybrid models more clearly, we carry out three more in-depth experiments on attention distribution for the RoPE-NoPE hybrids.

1. **Hit Rate Experiment**. Input the prompt of the 4k SK1 task in RULER [44], and calculate the probability that the top-k attention score in RoPE or NoPE attention hits the position to be retrieved for the query tokens at the end of the input. This experiment determines whether RoPE or NoPE is more responsible for direct information localization.
2. **Retrieval Head Experiment**. Follow the definition of the retrieval head and streaming head in DuoAttention [82] and compare the retrieval scores in RoPE attention and NoPE attention. DuoAttention's retrieval head refers to an attention head that, during inference, can retain not only the initial sink and the final local portion, in contrast to the streaming head. This experiment determines whether RoPE or NoPE attention reads distant content more frequently.
3. **Attention Entropy Experiment**: Count the attention entropy of RoPE and NoPE attention on the PG19 dataset [41]. We normalize the entropy between 0 and 1 by dividing by the maximum possible entropy value, the logarithm of the context length. This experiment determines whether RoPE or NoPE attention has a broader attention range.

Since retrieval scores are concentrated near 0 within $(0, 1)$, we divide them into three bins: high $[10^{-3}, 1)$, medium $[10^{-5}, 10^{-3})$, and low $(0, 10^{-5})$, and report each proportion. The results are shown in Figure 12, where NoPE and RoPE show the biggest gap in the hit-rate plot when $k = 64$. To better reflect the distinct roles of RoPE and NoPE attentions, we add a scatter plot where each point represents an attention head, using its attention entropy at 4k and its top-64 hit rate as its coordinates. In this **entropy-hit-rate diagram**, we also use different colors to distinguish the retrieval head and streaming head in RoPE and NoPE attention.

We first compare the RoPE-NoPE head-wise hybrid models with a 1:1 head mix. We find that **NoPE attention has a higher hit rate, a higher retrieval-head proportion, and a higher attention entropy, mainly occupying**



(a) Comparison within 1:1 RoPE-NoPE-HH. **(b)** Comparison among RoPE-NoPE-HH

Figure 12 The results of attention analysis within 1:1 RoPE-NoPE-HH and among RoPE-NoPE-HH with different mixing ratio. NoPE full attention shows a stronger tendency for coarse global information localization, given the higher attention score hit rate and larger attention entropy, compared with RoPE full attention in RoPE-NoPE hybrids.

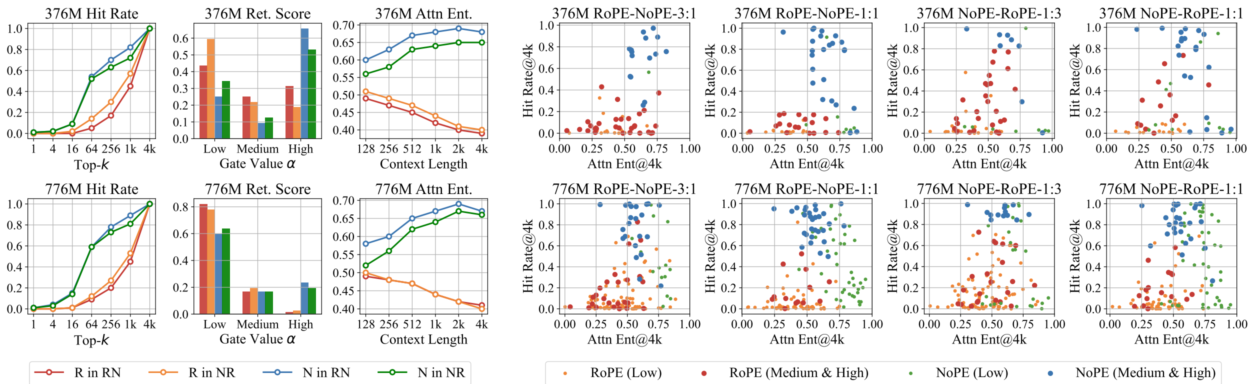
the strip area from the upper middle to the lower right in the entropy-hit-rate diagram. In contrast, **RoPE attention has a lower hit rate, a higher streaming-head proportion, and a lower attention entropy, mainly occupying the sector area extending outward from the lower-left corner** in the entropy-hit-rate diagram. This indicates that **NoPE mainly occupies coarse global localization** in the hybrid models, since it dominates the high-entropy region. However, the negative correlation between entropy and hit rate indicates that when NoPE achieves a medium or lower hit rate, its higher attention entropy may introduce more noise. That is why localization relying solely on NoPE is unreliable, and a higher NoPE ratio may imply worse performance.

This is even more evident in comparing RoPE-NoPE hybrids with different RoPE-NoPE ratios. As the NoPE ratio increases, the distribution of RoPE attentions in the entropy-hit-rate diagram gradually shrinks towards the lower-left corner. In contrast, NoPE attentions occupy the high-hit-rate region and extend to the low-hit-rate and high-entropy region simultaneously. This is just like a tidal lane, where traffic may travel in either direction, depending on certain conditions. Therefore, we name this redistribution, under the collaboration between position-biased attention and NoPE attention, the **Tidal Effect of Hybrid Position**, which is **Takeaway 4** in this paper: position-biased and NoPE attention heads occupy distinct regions in the entropy-hit-rate diagram, and their functional boundaries shift with the hybrid ratio.

Takeaway 4: Tidal Effect in Collaboration of Hybrid Position

Effective collaboration of hybrid position relies on a minority of high-hit-rate NoPE attention with coarse aggregation and a majority of low-entropy position-biased attention (i.e., RoPE attention and gated linear attention) for noise reduction. In the entropy-hit-rate diagram, position-biased attention occupies a sector region extending from the lower-left corner, while NoPE attention occupies a strip region from the upper middle to the lower right, and their boundaries shift with the hybrid ratio.

Notably, here we extend this effect from RoPE-NoPE hybrid modes to other hybrid models, including GLA-NoPE and GDN-NoPE hybrids. We will verify such an extension in Section 6. Here, we first verify the Tidal Effect in RoPE-NoPE layer-wise hybrid models. Considering the impact of layer order and functional differences at different depths, we conduct the validation experiment on 3:1 RoPE-NoPE, 1 NoPE layer after 3 RoPE layers, 1:1 RoPE-NoPE, 1 NoPE after 1 RoPE layer, 1:3 NoPE-RoPE, 3 RoPE layers after 1 NoPE layer, and 1:1 NoPE-RoPE, 1 RoPE after 1 NoPE layer. We compare the hit-rate curve, retrieval score distribution, and attention entropy curve in both 1:1 RoPE-NoPE (NR) and 1:1 NoPE-RoPE (RN), and present the entropy-hit-rate diagram for 4 hybrids in 376M and 776M model sizes in Figure 13. The partial orders in



(a) Comparison within 1:1 RoPE-NoPE-LH.

(b) Comparison among RoPE-NoPE-LH

Figure 13 The results of attention analysis within 1:1 RoPE-NoPE-LH and among RoPE-NoPE-LH with different mixing ratios. The tidal effect of hybrid position is still applicable in RoPE-NoPE layer-wise hybrid models.

hit rate, retrieval score, and attention entropy, and the distribution pattern between RoPE and NoPE attention are consistent with those in head-wise hybrid models. In addition, we find that attentions in upper layers tend to have higher hit rates and larger entropy for both NoPE and RoPE attentions.

To sum up, we clarify how RoPE and NoPE attention collaborate, extending Takeaway 1 to a more fine-grained level. Although NoPE attention enables training-free effective length extrapolation and a higher attention hit rate, its coarse global localization still requires collaboration with position-biased attention to maintain performance in long-context tasks as the distribution shifts with the hybrid ratio.

Takeaway 1: From Hybrid Attention to Hybrid Position (Complete Version)

Hybrid models improve long-context performance and length extrapolation through a hybrid position of NoPE and other position biases. Particularly,

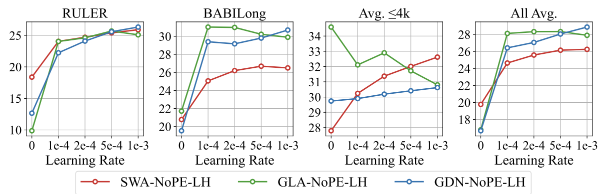
- On one hand, training-free effective length extrapolation can be achieved even for LLMs pretrained with full attention by limiting the attention scope of position bias.
- On the other hand, although NoPE attention consistently aggregates global information, downstream performance still relies on collaboration with other modules.

5 Discussion on SWA Hybrid

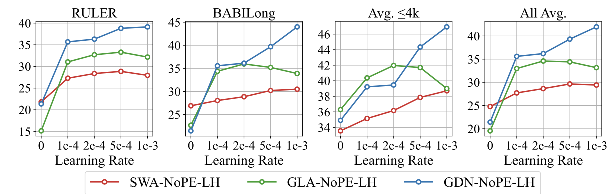
For SWA hybrids, we focus on why the seesaw effect appears during context extension, and why a model that extrapolates well from short-context pretraining can become less competitive after long-context continual pretraining. We first verify that this reversal is stable under different learning rates, then use perplexity curves and parameter change ratios to analyze the source of the problem. Finally, we improve context-extension performance by enlarging the window size in long-context continual pretraining and strengthening long-context dependency learning with LongCE [42].

5.1 Short-Context Learning Trap of SWA Hybrids

To verify that the seesaw effect is stable, we compare long-context continual pretraining results of SWA and LA hybrids under different peak learning rates, including 1e-4, 2e-4, 5e-4, and 1e-3, where 1e-3 is the default learning rate. We conduct this comparison at both 376M and 776M scales. The results are shown in Figure 14, and the point with a zero learning rate denotes the direct length extrapolation result before long-context



(a) Results of 376M models.



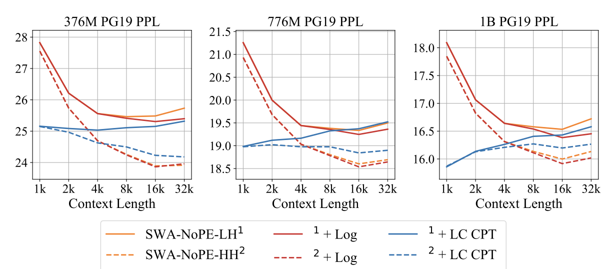
(b) Results of 776M models.

Figure 14 Comparison of long-context training performance between SWA-NoPE-LH, GLA-NoPE-LH, and GDN-NoPE-LH under different learning rates, where 0 learning rate means the performance before long-context continual pretraining.

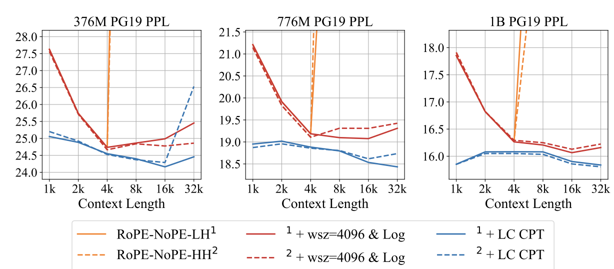
continual pretraining. The seesaw effect remains stable under different learning rates: after context extension, SWA-NoPE hybrids, especially on long-context tasks, consistently lag behind LA-NoPE hybrids.

We further verify the seesaw effect by observing the perplexity curves, and this reveals a potential reason behind the reversal in downstream task ranking. Figure 15 compares the perplexity curves of SWA-NoPE, RoPE-NoPE, GLA-NoPE, and GDN-NoPE hybrids before and after long-context continual pretraining. For most models, the perplexity curve before long-context training rises clearly beyond the pretraining context length. After long-context continual pretraining, the curve becomes much flatter or steadily decreases, indicating that the model can support longer contexts and use context to predict the next token better.

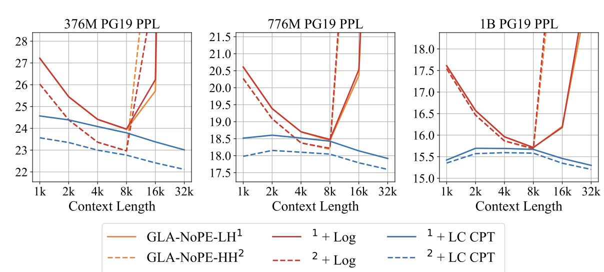
However, SWA-NoPE hybrids show the opposite behavior. Because they already have strong extrapolation ability, their perplexity curves keep decreasing within 32k before long-context continual pretraining. After long-context continual pretraining, especially for layer-wise SWA-NoPE hybrids, the perplexity curve instead shows an upward trend: **the loss on long-context positions decreases only slightly**, while **the loss on short-context positions drops abnormally strongly**, and this phenomenon becomes more obvious as the model size increases. This means that long-context dependencies do not help SWA-NoPE hybrids make better



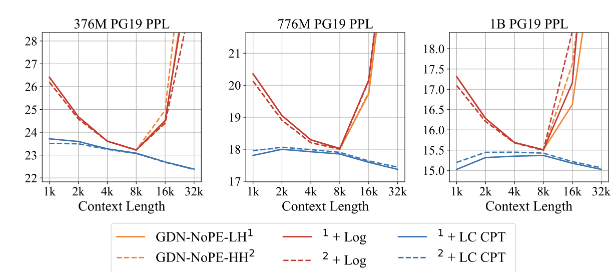
(a) Perplexity of SWA Hybrids.



(b) Perplexity on RoPE-NoPE hybrids.

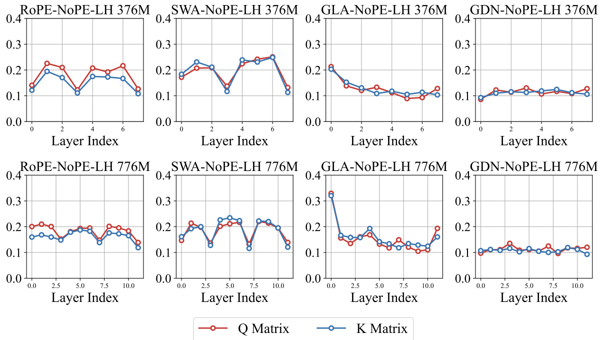


(c) Perplexity of GLA Hybrids.

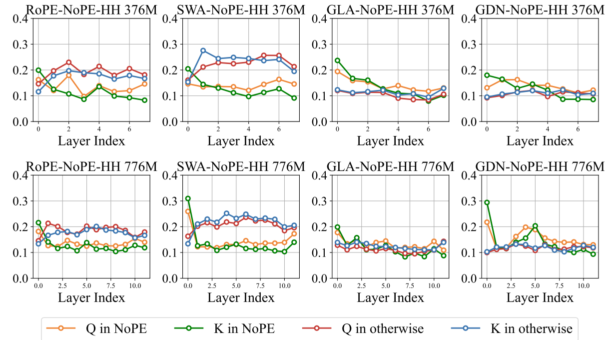


(d) Perplexity of GDN Hybrids.

Figure 15 Perplexity curves of different hybrid models under direct extrapolation, log-scale NoPE extrapolation (Log), and long-context continual pretraining (LC-CPT). The perplexity curve of SWA-NoPE-LH shows a remarkable increasing trend as the context length grows, different from other hybrid models.



(a) Comparison of layer-wise hybrid models.



(b) Comparison of head-wise hybrid models.

Figure 16 Comparison of the change rates of the $\boldsymbol{W}_Q$ and $\boldsymbol{W}_K$ matrices in attention modules across different hybrid models before and after long-context training. In layer-wise hybrids, 1 NoPE layer is inserted every 3 RoPE/SWA/GLA/GDN layers. In head-wise hybrids, each layer contains the $\boldsymbol{W}_Q$ and $\boldsymbol{W}_K$ matrices of NoPE as well as those of the other RoPE/SWA/GLA/GDN modules. Remarkably, in SWA-NoPE hybrids, the SWA modules receive anomalous emphasis during long-context training. However, their limited window size cannot support modeling longer contexts.

predictions. Long-context continual pretraining of SWA-NoPE hybrids focuses too much on short contexts, and as a result, LA-NoPE hybrids surpass SWA-NoPE hybrids after long-context continual pretraining. We refer to this phenomenon as the **Short-Context Learning Trap in Context Extension of SWA Hybrid**. This explains why the seesaw effect emerges during context extension of SWA hybrids and refines Takeaway 2.

Takeaway 2: Seesaw Effect in Context Extension of SWA Hybrid (Complete Version).

For the SWA-NoPE hybrid model, the advantage in long-context performance over the LA-NoPE hybrid is inverted after long-context continual pretraining, which is more remarkable in SWA-NoPE layer-wise hybrids, due to the **Short-Context Learning Trap in Context Extension of SWA Hybrid**.

To further verify the existence of short-context learning traps, we plot the parameter change ratios in Figure 16, comparing SWA-NoPE, RoPE-NoPE, GLA-NoPE, and GDN-NoPE hybrids. The x-axis represents different layers, and the y-axis represents the parameter change ratios of the $\boldsymbol{W}_Q$ and $\boldsymbol{W}_K$ matrices. Taking $\boldsymbol{W}_Q$ as an example, let $\boldsymbol{W}_Q$ and $\boldsymbol{W}'_Q$ denote its values before and after long-context continual pretraining. We define the change ratio as $\|\boldsymbol{W}'_Q - \boldsymbol{W}_Q\|_2 / \|\boldsymbol{W}_Q\|_2$. We first compute this ratio for each head within the same layer, and then average over heads. For head-wise hybrids, we separately aggregate NoPE heads and non-NoPE heads in each layer. For layer-wise hybrids, we do not make an additional distinction within each layer, and the non-NoPE layers and NoPE layers are interleaved at a 3:1 ratio.

We find that for GLA-NoPE and GDN-NoPE hybrids, the parameter change ratios of GLA/GDN and NoPE attention are close. In contrast, for SWA and RoPE attention, especially SWA, the parameter change ratio is significantly larger than that of NoPE attention in both layer-wise and head-wise hybrids. This suggests that **during long-context continual pretraining, SWA and RoPE attention receive much more optimization** in SWA-NoPE and RoPE-NoPE hybrids. For SWA, however, the window size remains fixed, so the range of context dependencies that it can model is still limited. Therefore, its abnormal focus on this bounded local branch is what drives SWA-NoPE hybrids into the short-context learning trap during context extension.

5.2 Short-Window Weariness and Long-Window Laziness

Motivated by Figure 16, we enlarge the SWA window size during long-context continual pretraining of layer-wise SWA-NoPE hybrids. The idea is to use the abnormal focus on the SWA branch to capture longer context dependencies. Specifically, we increase the window size from 128 to 256, 512, 1024, 2048, and 4096.

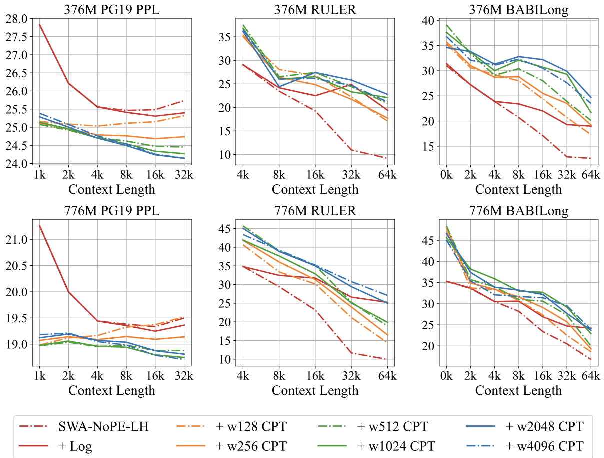


Figure 17 Perplexity curves of SWA hybrids. There is a clear short-context trap after long-context continual pretraining (LC-CPT), especially for layer-wise hybrids.

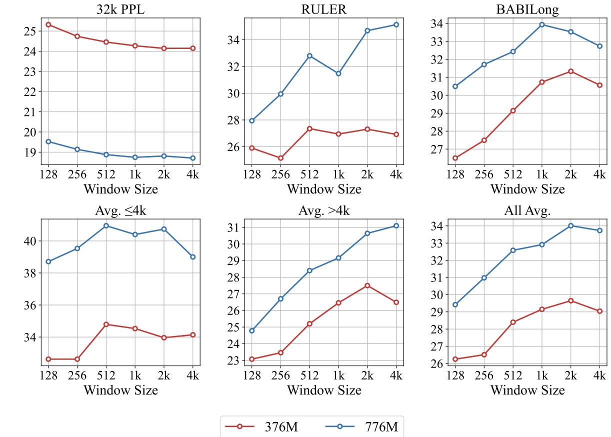


Figure 18 Comparison of long-context performance of SWA-NoPE-LH with different window sizes (wsz) in long-context continual pretraining.

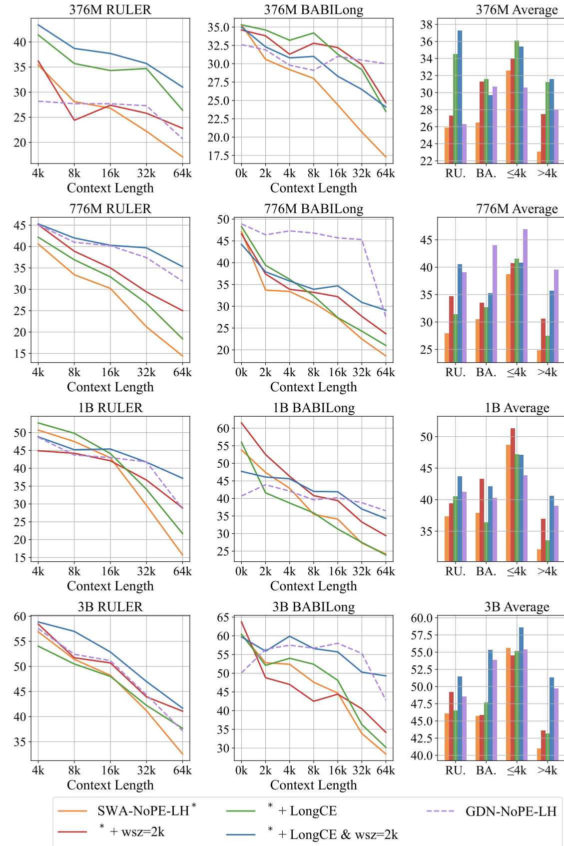
Following the scaling law of RoPE-based extrapolation [79], we also compute the rotary base corresponding to each enlarged window size when the initial window size is 128, and the original rotary base is 10000. Figure 17 and Figure 18 compare the PG19 perplexity, RULER, and BABILong performance of SWA hybrids after enlarging the window. The results show that increasing the window size indeed improves downstream long-context performance, and a window size of 2048 gives the best results at both 376M and 776M scales.

In addition, enlarging the window only exploits the short-context learning trap rather than emphasizing long-context dependencies directly. Therefore, we also use LongCE [42] as the training objective for layer-wise SWA-NoPE hybrids in long-context continual pretraining. As shown in Figure 19, both enlarging the window to 2048 and using LongCE improve the long-context performance of SWA-NoPE layer-wise hybrids, and the two methods can be combined. With both methods, SWA-NoPE-LH becomes competitive with GDN-NoPE-LH across different model sizes. Since the short-context trap in SWA hybrid long-context continual pretraining may not be limited to SWA-NoPE layer-wise hybrids, we further apply the same two strategies to SWA-RoPE layer-wise hybrids. Enlarging the window and using LongCE also improves the long-context training performance of SWA-RoPE-LH at different scales, making it close to GDN-RoPE-LH.

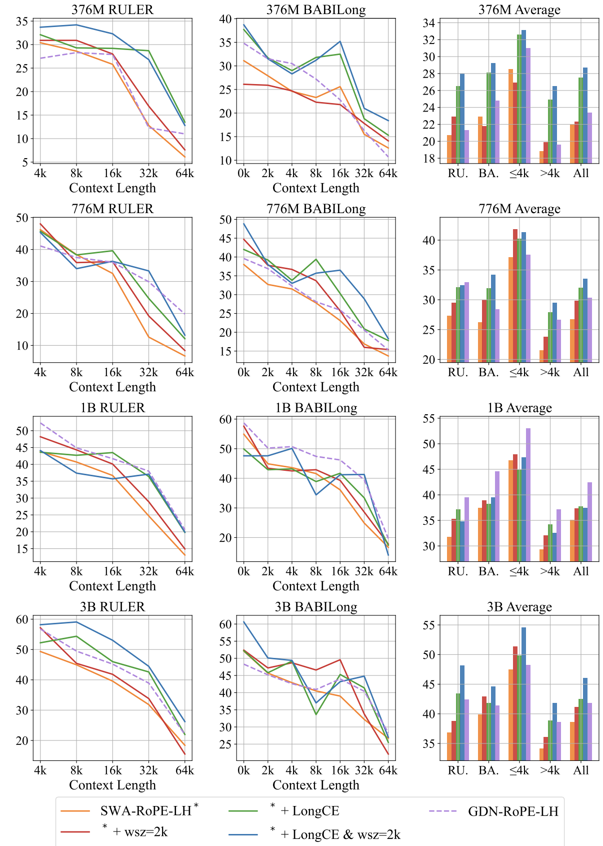
These results show that keeping a small SWA window during long-context continual pretraining harms context extension. **SWA hybrids suffer from the short-context learning trap and require enlarged windows as well as training strategies that emphasize long-context learning**. We call this phenomenon **Short-Window Weariness of SWA Hybrid**. At the same time, some previous works have reported that for SWA hybrids, shorter windows can be more advantageous than longer windows during the initial training stage, because they better activate the potential of NoPE full-attention layers [27, 34]; this phenomenon is sometimes called **Long-Window Laziness**. We argue that the two observations are not contradictory. The coexistence of short-window weariness and long-window laziness indicates that SWA hybrids may require fine-grained adjustment of positional bias across short-context and long-context training stages, just as RoPE full-attention models need to tune the rotary base and scaling factor in context extension [76, 77, 79, 83]. We summarize the coexistence of short-window weariness and long-window laziness as **Takeaway 5** in this paper.

Takeaway 5: Short-Window Weariness and Long-Window Laziness of SWA Hybrid

- In length extrapolation, SWA hybrids with short window size perform better.
- In context extension, SWA hybrids with sliding window extended to a larger size perform better.



(a) Results of SWA-NoPE-LH.



(b) Results of SWA-RoPE-LH.

Figure 19 Comparison of long-context performance of SWA-NoPE-LH after long-context continual pretraining enhanced by LongCE and extended window size. Both clearly weaken the seesaw effect and also contribute to SWA-RoPE-LH.

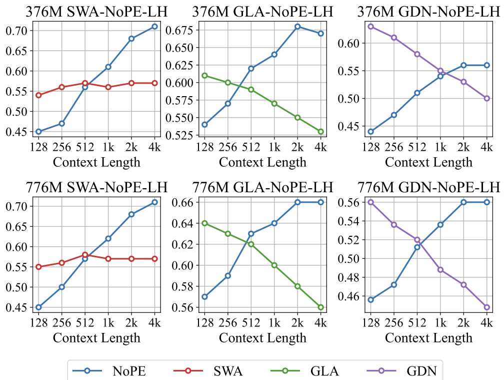
6 Discussion on LA Hybrid

For LA hybrids, we first ask how to analyze linear attention variants such as GLA and GDN, how to present their collaboration with NoPE attention, and whether this collaboration differs from that between RoPE attention and NoPE attention. Additionally, since GLA-NoPE and GDN-NoPE hybrids still underperform SWA hybrids in length extrapolation, we also need to consider how to improve their extrapolation performance.

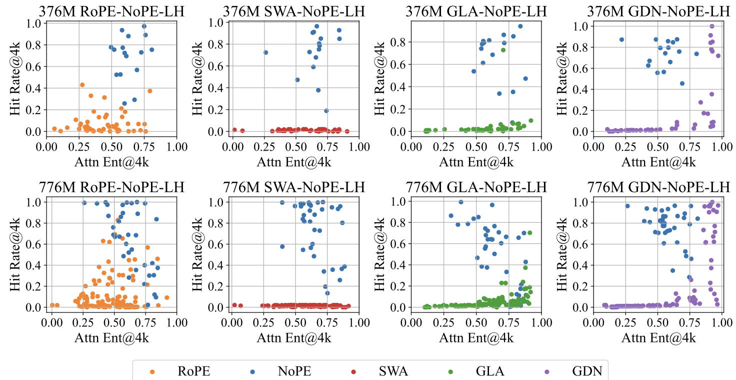
6.1 Extended Hybrid Position Analysis

The biggest difference between linear attention and softmax attention is that linear attention does not have a standard attention distribution, and its attention scores are not necessarily non-negative. Therefore, we cannot compute the attention entropy directly. Of course, one could apply softmax to the attention scores of linear attention and obtain a distribution, but this is inconsistent with the original computation of linear attention. Before analyzing linear attention, we therefore need a method that extends concepts such as attention distribution and attention entropy to arbitrary attention mechanisms, compatible with softmax attention and without extra operations on other attention algorithms.

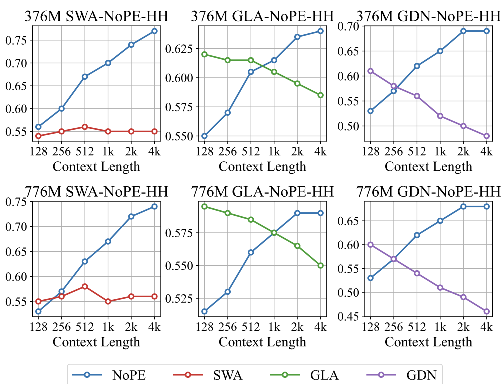
We note that the original attention distribution $\alpha_{t,s}$ is not only the softmax of the scaled inner product between $\boldsymbol{q}_t$ and $\boldsymbol{k}_s$, but also the combination weight of $\boldsymbol{v}_s$ when forming $\boldsymbol{o}_t$. From this perspective, $\alpha_{t,s}$ depends on the internal attention computation only in the forward pass. In the backward view, it corresponds to the



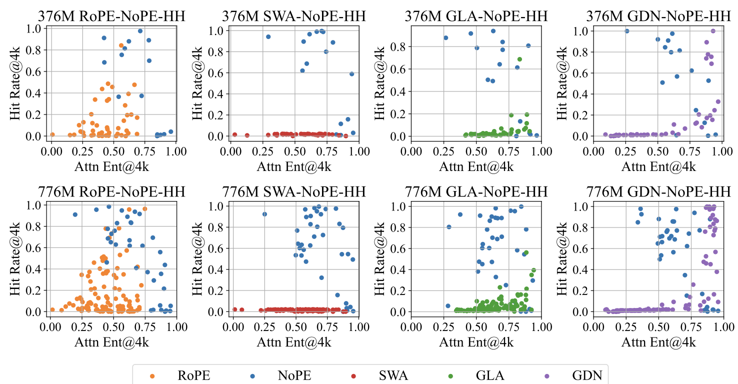
(a) Entropy comparison within NoPE-LH.



(b) Entropy-hit-rate scatter diagram comparison among NoPE-LH.



(c) Entropy comparison within NoPE-HH.



(d) Entropy-hit-rate scatter diagram comparison among NoPE-HH.

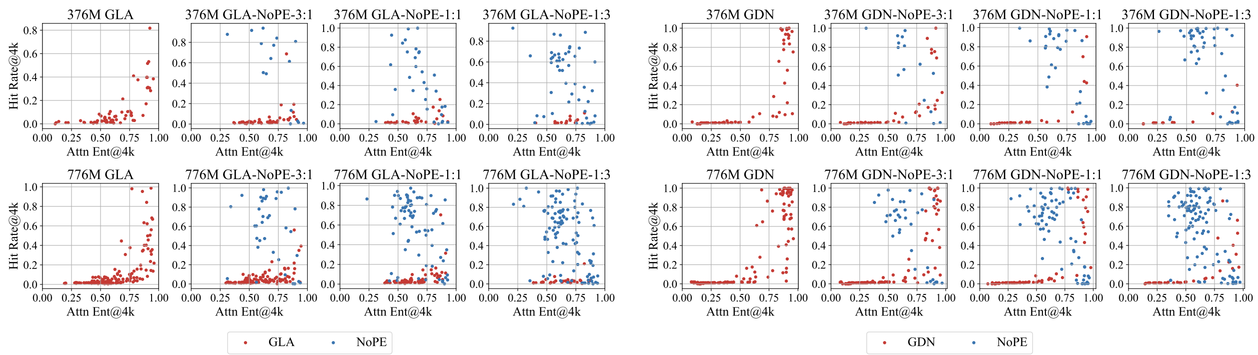
Figure 20 Attention analysis results among 3:1 RoPE-NoPE, SWA-NoPE, GLA-NoPE, and GDN-NoPE hybrid models, with extended attention entropy and hit rate. In the entropy-hit-rate diagram, NoPE attention still shows a stronger tendency to aggregate global information coarsely, with a higher attention hit rate and larger attention entropy. In contrast, SWA, GLA, and GDN show a similar distribution pattern to RoPE attention.

derivative of $\boldsymbol{o}_t$ with respect to $\boldsymbol{v}_s$, more precisely, the norm of the Jacobian matrix as shown in Equation 1.

$$\alpha_{s,t} = \operatorname{softmax}\left(\frac{\boldsymbol{q}_t \boldsymbol{k}_s^\top}{\sqrt{d_k}}\right), \quad \boldsymbol{o}_t = \sum_{s=0}^{t} \alpha_{s,t} \boldsymbol{v}_s, \quad \frac{\mathrm{d}\boldsymbol{o}_t}{\mathrm{d}\boldsymbol{v}_s} = \alpha_{s,t} \boldsymbol{I}, \quad \left\| \frac{\mathrm{d}\boldsymbol{o}_t}{\mathrm{d}\boldsymbol{v}_s} \right\|_2 = \alpha_{s,t}. \tag{1}$$

Since a matrix norm is non-negative, after normalization it yields a value in $[0, 1]$ and can be used as a distribution to compute attention entropy. For softmax attention, because $\alpha_{t,s}$ is already normalized, this extension does not change the original result. For linear attention, this extension does not modify the original attention computation. We call the resulting distribution the extended attention distribution, and call its entropy the extended attention entropy. By default, we also normalize this entropy by dividing by the logarithm of the context length. We use the spectral norm, namely the ℓ_2-norm, as the norm measure, compute extended attention entropy and top-k hit rate, and use them to discuss the characteristics of SWA, GLA, and GDN beyond RoPE attention. The results are shown in Figure 20 and Figure 21.

Figure 20 first compares the characteristics of different attention modules in 3:1 layer-wise SWA-NoPE, RoPE-NoPE, GLA-NoPE, and GDN-NoPE hybrids. Because SWA always keeps a fixed attention range, its attention entropy changes only slightly as the context length increases. Correspondingly, SWA also struggles to hit key information in the previous context, and its points stay close to the x-axis in the entropy-hit-rate diagram. NoPE attention behaves similarly to Figure 12: its entropy still increases with context length,



(a) Comparison among GLA-NoPE-HH

(b) Comparison among GDN-NoPE-HH

Figure 21 Attention analysis results among GLA-NoPE-HH and GDN-NoPE-HH with different hybrid ratios. The tidal effect of hybrid position is still applicable in GLA-NoPE and GDN-NoPE head-wise hybrid models.

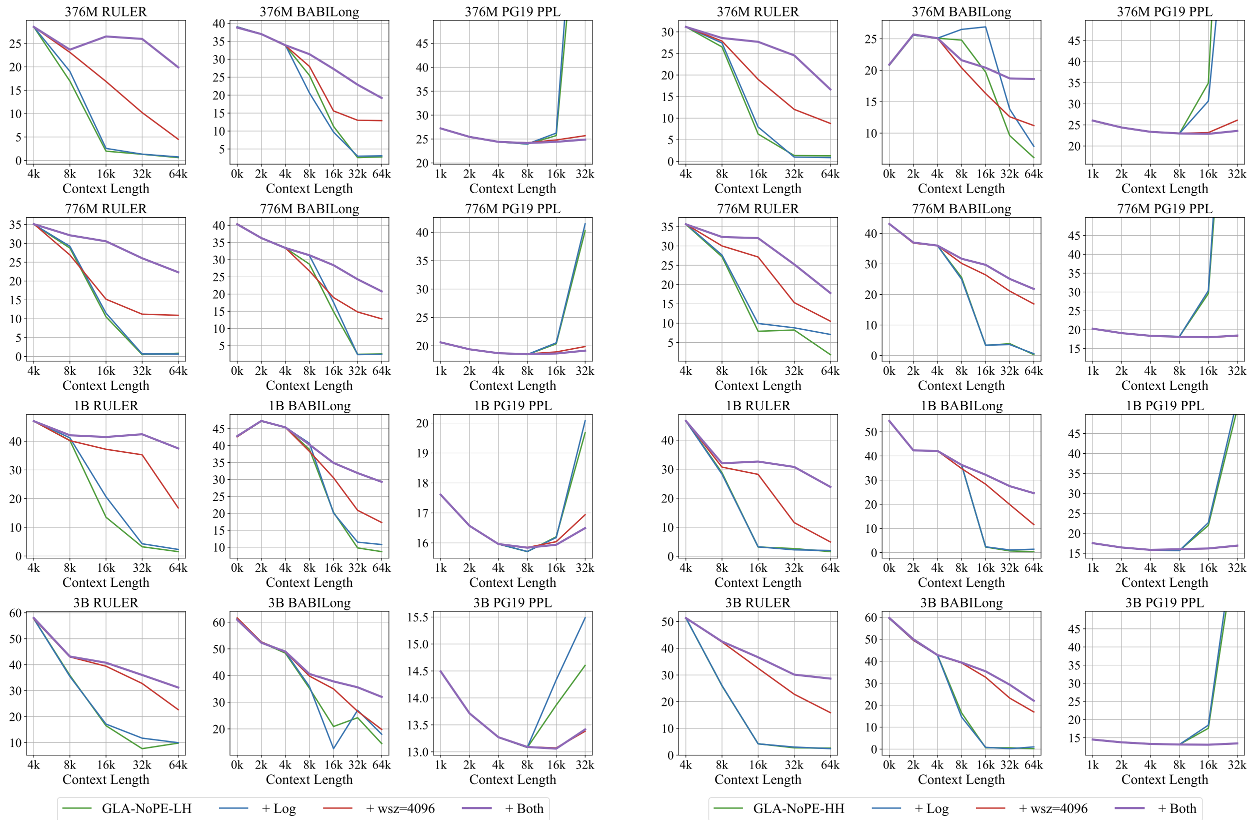
indicating a more uniform distribution and a coarser aggregation. In the entropy-hit-rate diagram, NoPE attention again occupies the strip region from the upper middle to the lower right. In contrast, **GLA and GDN behave more like the RoPE attention** in Figure 12: their entropy decreases as the context length increases, indicating that they perceive only a limited amount of information and become more concentrated at longer lengths. The difference is that GLA and GDN start from a higher-entropy region than RoPE and NoPE, which we believe is related to the absence of softmax in GLA and GDN: high-score positions are not explicitly amplified by softmax. Therefore, in the entropy-hit-rate diagram, GLA and GDN still occupy regions close to the lower-left area like RoPE attention, but overall shift toward the right, higher-entropy side.

We further compare head-wise GLA-NoPE and GDN-NoPE hybrids with different mixing ratios in Figure 21. Similar to Figure 12, as the NoPE ratio increases, the distribution of GLA or GDN in the entropy-hit-rate diagram gradually shrinks toward the lower-left region, while NoPE attention simultaneously occupies the high-hit-rate region. A difference from RoPE attention is that because the distributions of GLA and GDN are overall compressed toward the right, they also have points in the lower-right region with low hit rate and high entropy. This corresponds to the observation in Figure 21 that RoPE-NoPE hybrids are more sensitive to the increase of the NoPE ratio than LA-NoPE hybrids. Therefore, we answer the hypothesis left in Section 4, and extend the Tidal Effect of Hybrid Position from RoPE-NoPE hybrids to mixtures of NoPE attention and position-biased attention, including RoPE attention as well as gated linear attention such as GLA and GDN. This naturally leads to the next question: can we also use the extrapolation strategy of RoPE-NoPE hybrids [25, 28] to enhance the length extrapolation of GLA-NoPE and GDN-NoPE hybrids?

6.2 Length Extrapolation of LA Hybrid

In RoPE-NoPE hybrids, we achieve length extrapolation beyond traditional NTK interpolation by treating RoPE attention as a sliding-window attention module with a 4k window, while keeping NoPE attention globally aware and strengthening it in a log-scale correction [25, 28]. For LA-NoPE hybrids, we analogously propose **Sliding-Window Linear Attention**, which applies a sliding window to linear attention with gates. This may sound counterintuitive, because it appears to conflict with the recurrent nature of linear attention, but it can still be implemented. Without modifying FLA kernels [71], we use an approximation similar to sliding-window perplexity evaluation [36]. Suppose the window size is 4096 and the stride is 1024. We can use the original GLA or GDN operator to compute attention on chunks $[0, 4095], [1024, 5119], [2048, 6143], \cdots$ in parallel, and then extract the outputs at $[0, 4095], [4096, 5119], [5120, 6143], \cdots$ before stacking them back together. This implementation is independent of the internal details of a specific LA variant.

From the internal logic of linear attention, using GLA as an example, the original interaction between $\boldsymbol{q}_t$ and $\boldsymbol{k}_s$ is multiplied by $\prod_{i=s+1}^{t} \mathrm{Diag}(\boldsymbol{\alpha}_i)$. After applying Sliding-Window Linear Attention, the number of accumulated terms in this product is upper-bounded, implying that the relative-position signal is also



(a) Results on GLA-NoPE-LH. **(b)** Results on GLA-NoPE-HH.

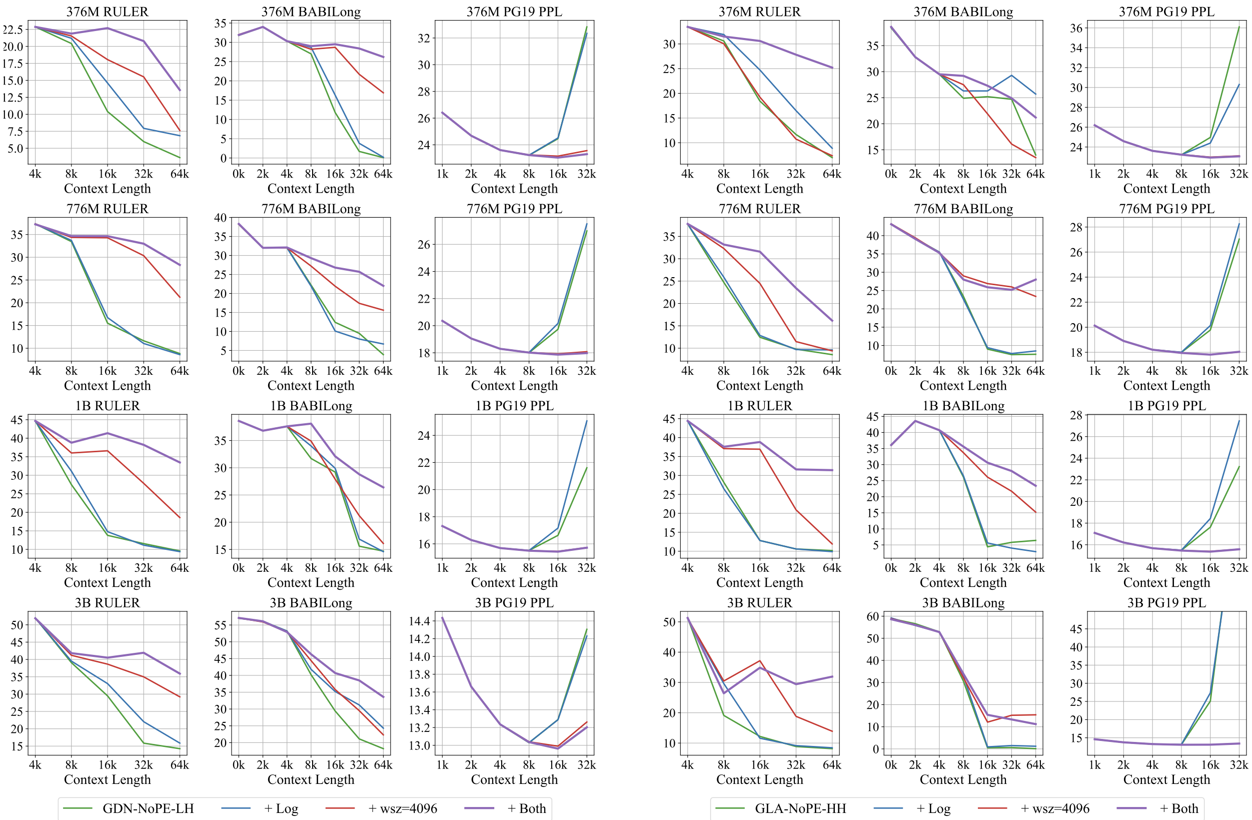
Figure 22 Comparison of length extrapolation in GLA-NoPE hybrid models. Applying a sliding window for GLA and a log scale for NoPE attention achieves the best length extrapolation performance.

restricted to the training length. As shown in Figures 22 and 23, GLA-NoPE and GDN-NoPE hybrids with Sliding-Window Linear Attention achieve clear improvements in length extrapolation. Moreover, directly adding log-scale correction to the NoPE attention in LA-NoPE hybrids brings only limited improvement, but after applying SWLA, log-scale correction becomes much more effective. This also implies that **even for linear attention with implicit positional bias, extrapolation should still restrict the relative-position range**. This gives **Takeaway 6** of this paper: in length extrapolation of hybrid-position models, position-biased attention should use a sliding window to restrict its relative-position information, while NoPE attention should strengthen its global perception. We call this phenomenon the **Matthew Effect of Length Extrapolation in Hybrid Position**, and abbreviate the **Extrapolation based on Matthew effect** as EME. This gives a simple yet effective extrapolation strategy, and on RoPE-NoPE hybrids it outperforms classical interpolation methods.

Takeaway 6: Matthew Effect of Length Extrapolation in Hybrid Position

A strong length extrapolation strategy for hybrid models is to apply a sliding window in position-biased attention (i.e., SWA for RoPE attention and SWLA for linear attention with gates), preserving only local perception, while increasing the attention temperature in NoPE attention to boost global aggregation.

In this paper, we have verified the effectiveness of EME on RoPE-NoPE, GLA-NoPE, and GDN-NoPE hybrids, and further shown on RoPE-NoPE hybrids that EME is better than traditional interpolation methods. Figure 24 summarizes the comparison among three hybrid architectures extrapolated with EME and SWA-



(a) Results on Layer-wise Hybrid Model (LH).

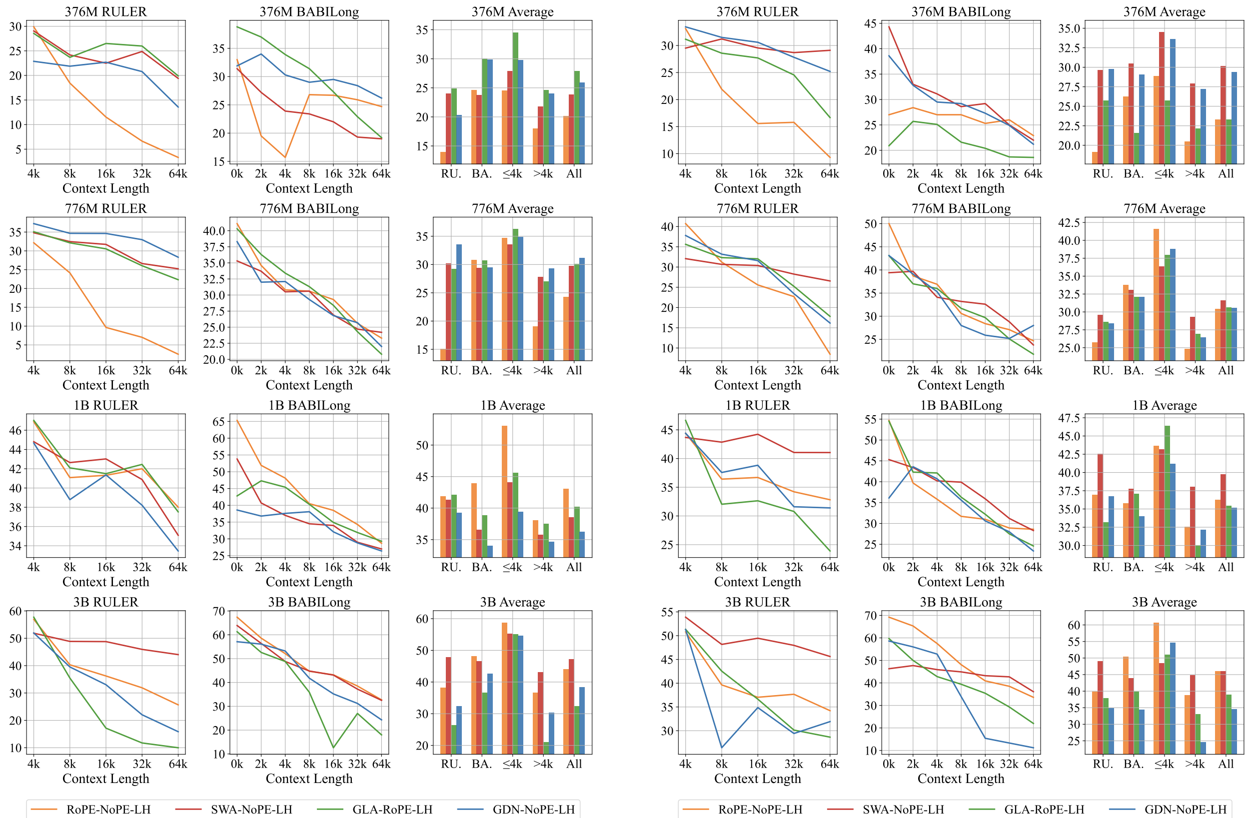
(b) Results on Head-wise Hybrid Model (HH).

Figure 23 Comparison of length extrapolation in GDN-NoPE hybrid models. Still, applying a sliding window for GDN and a log scale for NoPE attention achieves the best length extrapolation performance in most cases.

NoPE extrapolation. The four models show roughly comparable performance. We also highlight the results on the NIAH-SK1, SK2, and SK3 tasks in RULER, which are the most frequently reported tasks in hybrid model research. GLA-NoPE and GDN-NoPE with EME achieve 16× training-free length extrapolation while maintaining almost 100% retrieval accuracy on SK1. The results at the 776M scale are shown in Table 1, with more reported in Table 3. In fact, our finding is also consistent with reports from other work. For example, in Figure 5 of Chen et al. [55], the best extrapolating linear-attention hybrid is HypeNet-Lightning, and long-range decay is a key feature of Lightning Attention [84]. **Our approach based on SWLA further amplifies this property through the sliding window, breaks the traditional boundaries between SWA and LA, and is in principle compatible with arbitrary LA variants.**

6.3 Efficiency Comparison

Additionally, we include a brief inference efficiency comparison at the end of the paper. Across model sizes of 376M, 776M, 1B, and 3B, we compare traditional full-attention-only models with 3:1 layer-wise hybrid models that combine SWA, GLA, or GDN with NoPE full attention. We measure prefilling efficiency with Time to First Token (TTFT) and decoding efficiency with Time per Output Token (TPOT), both in microseconds. We evaluate memory efficiency based on peak allocated GPU memory and cache size, both measured in GB. As shown in Figure 25, hybrid models offer clear advantages as the model size and context length increase. Moreover, because the GLA and GDN variants store their recurrent states in higher FP32 precision, their resulting cache sizes are comparable to the cache overhead of the SWA hybrid with a window size of 128 and FP16 precision, so the corresponding curves overlap in the memory-efficiency comparison.



(a) Results on Layer-wise Hybrid Model (LH).

(b) Results on Head-wise Hybrid Model (HH).

Figure 24 Comparison of EME in RoPE-NoPE, GLA-NoPE, and GDN-NoPE hybrids as well as SWA-NoPE hybrids.

	NIAH-SK1					NIAH-SK2					NIAH-SK3				
	4k	8k	16k	32k	64k	4k	8k	16k	32k	64k	4k	8k	16k	32k	64k
776M Short															
GLA-NoPE-LH	99.0	95.0	80.0	0.0	0.0	99.0	69.0	0.0	0.0	0.0	90.0	88.0	0.0	0.0	0.0
+ Log	99.0	99.0	98.0	0.0	0.0	99.0	68.0	1.0	0.0	0.0	90.0	91.0	0.0	0.0	0.0
+ SWLA	99.0	**100.0**	**100.0**	97.0	97.0	99.0	74.0	5.0	0.0	0.0	90.0	56.0	21.0	8.0	3.0
+ SWLA & Log (EME)	99.0	**100.0**	**100.0**	96.0	**100.0**	99.0	92.0	87.0	72.0	59.0	90.0	**79.0**	**75.0**	**76.0**	**64.0**
GLA-NoPE-HH	**100.0**	**100.0**	68.0	82.0	0.0	**100.0**	71.0	0.0	0.0	0.0	65.0	45.0	0.0	0.0	0.0
+ Log	**100.0**	**100.0**	98.0	95.0	72.0	**100.0**	68.0	0.0	0.0	0.0	65.0	49.0	0.0	0.0	0.0
+ SWLA	**100.0**	**100.0**	**100.0**	92.0	75.0	**100.0**	**100.0**	82.0	32.0	1.0	65.0	62.0	44.0	9.0	1.0
+ SWLA & Log (EME)	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	97.0	66.0	31.0	65.0	68.0	65.0	62.0	34.0
GDN-NoPE-LH	**100.0**	**100.0**	**100.0**	**100.0**	99.0	99.0	98.0	2.0	0.0	0.0	62.0	71.0	5.0	0.0	0.0
+ Log	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	99.0	98.0	9.0	1.0	0.0	62.0	70.0	9.0	0.0	0.0
+ SWLA	**100.0**	**100.0**	**100.0**	**100.0**	99.0	99.0	97.0	96.0	92.0	53.0	62.0	64.0	61.0	56.0	31.0
+ SWLA & Log (EME)	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	99.0	97.0	**99.0**	**99.0**	**90.0**	62.0	67.0	64.0	66.0	53.0
GDN-NoPE-HH	**100.0**	**100.0**	**100.0**	99.0	92.0	**100.0**	95.0	2.0	0.0	0.0	**94.0**	38.0	0.0	0.0	0.0
+ Log	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	93.0	3.0	0.0	0.0	**94.0**	43.0	0.0	0.0	0.0
+ SWLA	**100.0**	99.0	**100.0**	95.0	92.0	**100.0**	99.0	64.0	4.0	0.0	**94.0**	68.0	31.0	5.0	2.0
+ SWLA & Log (EME)	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	97.0	72.0	35.0	**94.0**	71.0	65.0	47.0	16.0

Table 1 Extrapolation comparison of the 776M GLA/GDN-NoPE hybrid model on the NIAH-SK1, SK2, and SK3 tasks. Applying a sliding-window linear attention and a log scale for NoPE attention achieves the best length extrapolation.

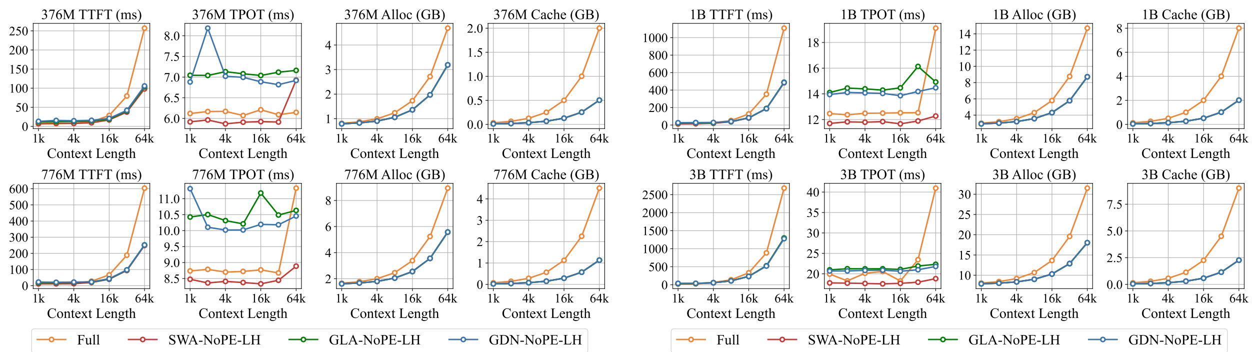


Figure 25 A brief efficiency comparison between the traditional full attention model and layer-wise hybrid models.

7 Conclusion

In this paper, we conduct extensive experiments to validate the effectiveness of the hybrid model in length extrapolation and context extension. From the perspective of hybrid position in NoPE attention versus other position-biased attention, we propose the Tidal Effect in collaboration and the Matthew Effect in extrapolation. We also discover the Seesaw Effect in context extension of the sliding-window attention hybrid models versus linear attention hybrid models. For SWA hybrids, we enlarge the window during extended training to overcome its short-context learning trap. For LA hybrids, we propose the Sliding-Window Linear Attention, achieving 16× training-free length extrapolation while maintaining 100% retrieval accuracy.

Limitations

This paper gives a mechanistic analysis of long-context hybrid models, with a broad loop of training, validation, observation, and induction. Even so, several directions remain open.

- **Validation scope**. Our current validation pipeline is focused on context extension and length extrapolation in the pretraining stage. We still need to extend it to post-training [31] and multimodal, especially vision-language [6, 85] models. On that basis, we should also extend our evaluation to complex reasoning, agentic tasks, multimodal retrieval, and multimodal understanding.
- **Architecture scope**. Our analysis focuses on RoPE/NoPE hybrids, SWA hybrids, and LA hybrids. We should apply the same mechanistic lens to other efficient hybrid families, such as sparse-attention [9, 37, 46] and compressed-attention [19, 40] models, but we leave this broader comparison for future work.
- **Mechanism scope**. Even within the hybrid families studied here, we only probe a subset of position and attention design choices. We will also further our study by analyzing techniques such as sink bias [12, 17, 86], partial RoPE [16, 18], gated attention variants [16, 87], and ablation of short convolution [33, 47, 67].

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Appendix

A Appendix

The configurations of models pretrained in this paper are summarized in the following table. Our models use the same tokenizer as the Llama 3 Series [4, 88, 89].

	376M	776M	1B	3B
Hidden Size	1024	1536	2048	3072
Intermediate Size	3584	5376	7168	10752
Num Layer	8	12	16	24
Num Attn Head	8	12	16	24
Num KV Head	8	12	16	24
Vocab Size	128256	128256	128256	128256

Table 2 The hyperparameters of different model sizes.

	NIAH-SK1					NIAH-SK2					NIAH-SK3				
	4k	8k	16k	32k	64k	4k	8k	16k	32k	64k	4k	8k	16k	32k	64k
376M Short															
GLA-NoPE-LH	100.0	0.0	0.0	0.0	0.0	100.0	88.0	0.0	0.0	0.0	39.0	35.0	0.0	0.0	0.0
+ Log	100.0	0.0	0.0	0.0	0.0	100.0	94.0	3.0	0.0	0.0	39.0	51.0	0.0	0.0	0.0
+ SWLA	100.0	70.0	79.0	75.0	46.0	100.0	94.0	57.0	9.0	0.0	39.0	34.0	17.0	16.0	0.0
+ SWLA & Log	100.0	70.0	81.0	84.0	84.0	100.0	99.0	97.0	92.0	68.0	39.0	34.0	52.0	70.0	37.0
GLA-NoPE-HH	100.0	100.0	75.0	0.0	0.0	100.0	98.0	0.0	0.0	0.0	96.0	75.0	0.0	0.0	0.0
+ Log	100.0	100.0	92.0	1.0	0.0	100.0	100.0	0.0	0.0	0.0	96.0	84.0	0.0	0.0	0.0
+ SWLA	100.0	100.0	100.0	100.0	93.0	100.0	98.0	54.0	15.0	5.0	96.0	69.0	29.0	6.0	1.0
+ SWLA & Log	100.0	100.0	100.0	100.0	100.0	100.0	99.0	98.0	91.0	49.0	96.0	81.0	89.0	77.0	37.0
GDN-NoPE-LH	99.0	97.0	88.0	67.0	43.0	75.0	28.0	1.0	0.0	0.0	45.0	82.0	13.0	0.0	0.0
+ Log	99.0	98.0	94.0	91.0	84.0	75.0	37.0	16.0	0.0	0.0	45.0	76.0	25.0	0.0	0.0
+ SWLA	99.0	97.0	82.0	62.0	40.0	75.0	57.0	54.0	31.0	12.0	45.0	65.0	61.0	74.0	23.0
+ SWLA & Log	99.0	97.0	90.0	88.0	79.0	75.0	59.0	71.0	54.0	33.0	45.0	63.0	68.0	71.0	22.0
GDN-NoPE-HH	100.0	98.0	96.0	89.0	79.0	99.0	94.0	37.0	16.0	0.0	99.0	97.0	37.0	2.0	2.0
+ Log	100.0	100.0	100.0	100.0	100.0	99.0	99.0	60.0	30.0	1.0	99.0	98.0	57.0	12.0	2.0
+ SWLA	100.0	100.0	100.0	90.0	82.0	99.0	95.0	20.0	0.0	0.0	99.0	88.0	60.0	28.0	2.0
+ SWLA & Log	100.0	100.0	100.0	100.0	100.0	99.0	99.0	91.0	76.0	71.0	99.0	96.0	93.0	91.0	83.0
1B Short															
GLA-NoPE-LH	100.0	98.0	6.0	0.0	0.0	100.0	100.0	48.0	2.0	0.0	91.0	82.0	8.0	0.0	0.0
+ Log	100.0	98.0	23.0	0.0	0.0	100.0	100.0	77.0	5.0	1.0	91.0	89.0	22.0	0.0	0.0
+ SWLA	100.0	100.0	100.0	100.0	100.0	100.0	100.0	82.0	85.0	33.0	91.0	78.0	59.0	52.0	1.0
+ SWLA & Log	100.0	100.0	100.0	100.0	100.0	100.0	100.0	98.0	100.0	99.0	91.0	85.0	73.0	73.0	71.0
GLA-NoPE-HH	96.0	24.0	0.0	0.0	0.0	100.0	100.0	0.0	0.0	0.0	92.0	80.0	0.0	0.0	0.0
+ Log	96.0	22.0	0.0	0.0	0.0	100.0	100.0	0.0	0.0	0.0	92.0	77.0	0.0	0.0	0.0
+ SWLA	96.0	35.0	20.0	6.0	1.0	100.0	100.0	99.0	10.0	0.0	92.0	85.0	75.0	46.0	0.0
+ SWLA & Log	96.0	47.0	59.0	77.0	64.0	100.0	100.0	100.0	99.0	68.0	92.0	86.0	81.0	80.0	66.0
GDN-NoPE-LH	100.0	100.0	100.0	100.0	100.0	100.0	69.0	0.0	0.0	0.0	76.0	47.0	0.0	0.0	0.0
+ Log	100.0	100.0	100.0	100.0	100.0	100.0	90.0	4.0	0.0	0.0	76.0	56.0	0.0	0.0	0.0
+ SWLA	100.0	100.0	100.0	100.0	100.0	100.0	91.0	93.0	73.0	32.0	76.0	62.0	60.0	44.0	12.0
+ SWLA & Log	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	96.0	76.0	66.0	76.0	69.0	64.0
GDN-NoPE-HH	100.0	100.0	100.0	100.0	100.0	100.0	97.0	0.0	0.0	0.0	69.0	17.0	0.0	0.0	0.0
+ Log	100.0	100.0	100.0	100.0	100.0	100.0	95.0	0.0	0.0	0.0	69.0	7.0	0.0	0.0	0.0
+ SWLA	100.0	100.0	100.0	100.0	97.0	100.0	100.0	97.0	36.0	5.0	69.0	62.0	55.0	19.0	9.0
+ SWLA & Log	100.0	100.0	100.0	100.0	100.0	100.0	100.0	99.0	91.0	96.0	69.0	63.0	61.0	56.0	55.0
3B Short															
GLA-NoPE-LH	100.0	46.0	10.0	33.0	62.0	100.0	98.0	54.0	3.0	0.0	100.0	82.0	25.0	0.0	0.0
+ Log	100.0	50.0	12.0	21.0	39.0	100.0	97.0	51.0	34.0	20.0	100.0	85.0	28.0	12.0	7.0
+ SWLA	100.0	74.0	81.0	71.0	69.0	100.0	100.0	100.0	94.0	53.0	100.0	98.0	82.0	75.0	38.0
+ SWLA & Log	100.0	75.0	67.0	56.0	64.0	100.0	100.0	100.0	99.0	95.0	100.0	99.0	94.0	85.0	69.0
GLA-NoPE-HH	88.0	3.0	0.0	0.0	0.0	100.0	93.0	0.0	0.0	0.0	100.0	41.0	0.0	0.0	0.0
+ Log	88.0	13.0	0.0	0.0	0.0	100.0	80.0	0.0	0.0	0.0	100.0	37.0	0.0	0.0	0.0
+ SWLA	88.0	41.0	15.0	1.0	0.0	100.0	100.0	100.0	86.0	59.0	100.0	92.0	62.0	54.0	26.0
+ SWLA & Log	88.0	44.0	20.0	6.0	5.0	100.0	100.0	100.0	100.0	100.0	100.0	95.0	76.0	69.0	73.0
GDN-NoPE-LH	100.0	100.0	100.0	100.0	100.0	100.0	100.0	88.0	4.0	0.0	95.0	64.0	15.0	0.0	0.0
+ Log	100.0	100.0	100.0	100.0	100.0	100.0	100.0	99.0	43.0	14.0	95.0	65.0	26.0	9.0	0.0
+ SWLA	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	92.0	90.0	95.0	75.0	66.0	56.0	33.0
+ SWLA & Log	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	98.0	97.0	95.0	80.0	69.0	68.0	66.0
GDN-NoPE-HH	100.0	100.0	100.0	99.0	99.0	100.0	6.0	0.0	0.0	0.0	96.0	60.0	0.0	0.0	0.0
+ Log	100.0	100.0	100.0	100.0	100.0	100.0	74.0	0.0	0.0	0.0	96.0	58.0	0.0	0.0	0.0
+ SWLA	100.0	100.0	100.0	100.0	96.0	100.0	78.0	100.0	22.0	10.0	96.0	55.0	60.0	21.0	4.0
+ SWLA & Log	100.0	100.0	100.0	100.0	100.0	100.0	39.0	100.0	98.0	99.0	96.0	57.0	58.0	42.0	63.0

Table 3 Extrapolation comparison of the GLA/GDN-NoPE hybrid model on the NIAH-SK1, SK2, and SK3 tasks. Applying a sliding-window linear attention and a log scale for NoPE attention achieves the best length extrapolation.